

h1 CHEF-KOCH GamingTweaks

h3 DirectX Control Panel

Some old(er) Games (and Emulators) might suffering from and outdated Engine which aren't compatible anymore with the latest Windows 10 DirectX changes, the [DirectX Control Panel](http://www.sephiroth-j.de/1/downloads/diverses.php#windows_dxcpl) (http://www.sephiroth-j.de/1/downloads/diverses.php#windows_dxcpl) utility addresses some of those engine related problems which means you can for example re-enable Hardware Acceleration functions on them.

Another way (without utility) would to do this manually for every Game via Registry but this is in most of the cases too complicated and not necessary since the utility is freeware and small in size it should be no problem to dl and use it, in most of the cases the program still does it's job.

h3 DirectX 11 vs 12 and FPS drops/stutters

According to this [post](https://linustechtips.com/main/topic/564099-fresh-problems-for-nvidia-users-running-directx-12/) (<https://linustechtips.com/main/topic/564099-fresh-problems-for-nvidia-users-running-directx-12/>) and [Computerbase benchmark](https://www.computerbase.de/2019-03/hitman-2-directx-12/) (<https://www.computerbase.de/2019-03/hitman-2-directx-12/>) most (if not all) modern D3D12 games (with nVidia hardware) have a regression which means you lose like *5 ~ 10%* in performance once you [DX12](https://www.extremetech.com/extreme/207598-demystifying-directx-12-support-what-amd-intel-and-nvidia-do-and-dont-deliver) (<https://www.extremetech.com/extreme/207598-demystifying-directx-12-support-what-amd-intel-and-nvidia-do-and-dont-deliver>). This is not fixable and a problem on nVidia side (AMD is not affected also not with Vulkan API based games e.g. Doom 2016). nVidia itself mention '[Do's and Dont's](https://developer.nvidia.com/dx12-dos-and-donts)' (<https://developer.nvidia.com/dx12-dos-and-donts>) but this doesn't help the enduser since this is depending on how the developer implements this suggestions in their game(s).

Recommendations:

- Switch back to DX11 or try to play Vulkan API based games (e.g. DOOM 2016).
- Cap the FPS (FPS cap [no matter which hardware] **is a must if you stream or record the game!**) The reason is that capturing programs like OBS Studio/XSplit etc trying to get as much as possible on resources for their own progress, overall it's all about resource management.
- Try to contact the developer and give him as much as details as you can provide, the best way would be to address this in a patch
- Make sure you use the last Windows 10 version, since older versions might suffer from additional DX problems
- Play (or try to switch [if possible]) Vulkan API based games such as Doom 2016 (this game runs the same on all hardware as proven [here](https://www.youtube.com/watch?v=1hnI6KGTfjI) (<https://www.youtube.com/watch?v=1hnI6KGTfjI>)). Personally it's beyond me why OBS Studio and several popular games does not supporting Vulkan API because it's known to run better than DirectX. *I assume it has something to do with exclusive Microsoft deals to keep this alive* or due to backward compatibility.

h2 Drivers (general advice)

- Try to uninstall every drivers which you won't need (e.g. on OEM systems [pre-installed drivers]) via Control Panel\All Control Panel Items\Programs and Features, say no if they ask you to reboot the OS. Make sure you reboot into "Safe Mode" to install important drivers in order to get full access, this is not mandatory but it might solve to remove all leftovers (in case you have some upgrade/install) issue.
- Disable unused audio devices (like ATI HDMI) in Device Manager (which are most of the time not needed)
- Don't use utilities like [Intel Driver Update Utility](https://www.intel.com/content/www/us/en/support/topics/idsa-faq.html) (<https://www.intel.com/content/www/us/en/support/topics/idsa-faq.html>) because it runs all the time in the background, submits telemetry and doesn't have any benefit over a *traditional* driver page like e.g. [Station-Drivers](https://www.station-drivers.com/index.php?lang=en) (<https://www.station-drivers.com/index.php?lang=en>).
- Do not install "Xtreme-G Modded Drivers", the promises the modders makes are ridiculous and there is no evidence that this holds what it promises.

h2 Chipset driver update myth

You [do not need to install any chipset driver unless you see a yellow triangle](https://www.win-raid.com/t895f42-Intel-Chipset-Device-quot-Diverses-quot-INF-files.html) (<https://www.win-raid.com/t895f42-Intel-Chipset-Device-quot-Diverses-quot-INF-files.html>) in device manager. Windows Update usually then kicks in and offers you the correct version. There is (*in normal situations*) no benefit in using an updated chipset drivers because if you [inspect the changelog](https://downloadmirror.intel.com/29227/eng/10.1.18228.8176_3.1.7.118_public.pdf) (https://downloadmirror.intel.com/29227/eng/10.1.18228.8176_3.1.7.118_public.pdf) you will notice that newer chipset driver versions often only adding new products they typically never "fix" something.

h4 Best practice

- If you see no yellow triangle in Windows device manager (*unknown device or error codes*) do not install any chipset driver.
- If Windows Update offers you the driver then you're good to go, just install it and reboot.
- Monitor the changelogs from AMD and Intel for platform specific changes, if you see new chipset driver versions just do not blindly download them, instead check if the changelog mention some improvements (*they never do*).

h2 nVidia Driver update procedure

- Download the latest official driver of the [clean driver](https://github.com/CHEF-KOCH/nVidia-modded-Inf) (https://github.com/CHEF-KOCH/nVidia-modded-Inf).
- Download/use DDU under Windows safe mode and then restart the PC after the cleaning is done.
- Install the new driver and then restart your PC.
- (optional) Activate **MSI mode** and restart your PC. MSI Mode needs to be set again after each new driver change!
- (optional) Configure via nVidia Inspector e.g. 3D Global settings (keep in mind you can import/export those settings!), MAXPRERENDER FPS = 1 + Anisotropic Filtering Sample Optimization ON, Texture filtering Quality = High Performance + Power management mode = Prefer Max Performance
- Configure 3D Game profile settings according to your own needs. E.g. Fast sync on for R6 Siege with an 120fps limiter for consistent and tear free gameplay.
- Use [Intelligent standby list cleaner \(ISLC\)](https://www.wagnardsoft.com/forums/viewtopic.php?t=1256) (https://www.wagnardsoft.com/forums/viewtopic.php?t=1256), [Timer Resolution](https://cms.lucashale.com/timer-resolution/) (https://cms.lucashale.com/timer-resolution/) or [Timer Tool](https://github.com/tebjan/TimerTool) (https://github.com/tebjan/TimerTool) and set the value to **0.499ms** (0.50ms) in order to reduce the overall latency. *ISLC is the all-in-one solution.*
- Make your own game profiles and put the fps limiters for your games using RivaTuner Statistics Server (RTSS) provided along with MSI Afterburner according to your systems GFX and CPU power.
- Evaluate your settings per game by testing each title you play so you can make further adjustments in case something is not working properly!
- (optional) Set **SILK Smoothness** to **OFF** for multiplayer FPS Shooters only, which might improve it's responsiveness.
- Set **Power management** to **Adaptive**.

h2 AMD Driver update procedure

- Download the latest driver from the official AMD page.
- Download/use DDU under Windows safe mode and then restart the PC after the cleaning is done.
- There are no additional tweaks (needed [?]) - if you found some, let me know or create a PR)

h2 nVidia and Ansel

[Ansel](https://www.geforce.com/hardware/technology/ansel) (https://www.geforce.com/hardware/technology/ansel) is (more or less) deprecated and already partially removed from the current nVidia drivers (only backward compatibility is still given). There are several games like WatchDogs or [Conan Exile](https://steamcommunity.com/app/440900/discussions/0/133256959371919651/) (https://steamcommunity.com/app/440900/discussions/0/133256959371919651/) which suffering from FPS drops once Ansel if its enabled.

If you have FPS drops make sure you disable Ansel the default **keybind to activate/disable Ansel is Alt+F2** which disables Ansel globally. Alternative manually open the file **NvCameraConfiguration.exe** under **C:\Program Files\NVIDIA Corporation\Ansel\Tools**.

Per Game via nVidia Inspector:

Scroll down in nVidia Inspector until you see the **GLOBAL_DRIVER_PROFILE** option you will see in this category an flag **ENABLE_ANSEL_ON** which you can toggle off. If there is no such entry it means that the installed driver doesn't support/include Ansel or that the Game itself doesn't use Ansel.

h4 Windows 10 (only) Upgrade drivers from legacy to UWD/MWD

[Modern Windows Drivers \(MWDs\) or also known as Universal Windows Drivers \(UWDs\)](https://docs.microsoft.com/en-us/windows-hardware/drivers/develop/getting-started-with-universal-drivers) drivers replacing the old driver scheme in [Windows](https://docs.microsoft.com/en-us/windows-hardware/drivers/develop/getting-started-with-universal-drivers) (https://docs.microsoft.com/en-us/windows-hardware/drivers/develop/getting-started-with-universal-drivers), this basically means for the end-user smaller driver packages and it includes more OEMs (universal).

- Remove the old „legacy“ driver via Apps.
- Reboot the OS.
- (optional) Check leftovers via [Driver Store Explorer](https://github.com/lostindark/DriverStoreExplorer) (https://github.com/lostindark/DriverStoreExplorer), because MS often keeps (as backup) the old driver in the Driver Store.
- Install the new UWD driver via setup.exe.
- Reboot
- **Important:** Do not install the new UWD via [.INF or Have-a-Disk method](https://www.computerhope.com/issues/ch000834.htm#have-disk) (https://www.computerhope.com/issues/ch000834.htm#have-disk) because this bypasses the setup.exe internal install method and can causes additional problems, crashes, performance issues etc.

h3 LatencyMon and how you correctly interpret the results

First of all [LatencyMon](https://www.resplendence.com/latencymon) (https://www.resplendence.com/latencymon) is a program for "power users" and not something for beginners but I try to explain it in some easy understandable steps which everyone can follow. Microsoft own utility is called [perfmon] ([https://docs.microsoft.com/en-us/previous-versions/windows/it-pro/windows-server-2008-R2-and-2008/cc749154\(v=ws.11\)](https://docs.microsoft.com/en-us/previous-versions/windows/it-pro/windows-server-2008-R2-and-2008/cc749154(v=ws.11))) (https://docs.microsoft.com/en-us/previous-versions/windows/it-pro/windows-server-2008-R2-and-2008/cc749154(v=ws.11)) or [xperf](https://docs.microsoft.com/en-us/windows-hardware/test/wpt/) (https://docs.microsoft.com/en-us/windows-hardware/test/wpt/) which also can read-out several things LatencyMon does.

h3 Warning

LatencyMon is no guarantee to find every OS/Bios or in-game related problem! Even if everything is in the green and you don't see any spikes you still can get FPS drops or other gaming issues.

LatencyMon checks:

- [Buffer underruns](https://en.wikipedia.org/wiki/Buffer_underrun) (https://en.wikipedia.org/wiki/Buffer_underrun) by measuring [kernel timer latencies](https://www.osr.com/nt-insider/2014-issue3/windows-real-time/) (https://www.osr.com/nt-insider/2014-issue3/windows-real-time/)
- Deferred Procedure Call [DPC](https://docs.microsoft.com/en-us/dotnet/api/system.windows.threading.dispatcher?view=netframework-4.8) (https://docs.microsoft.com/en-us/dotnet/api/system.windows.threading.dispatcher?view=netframework-4.8) and interrupt service routine [ISR](https://en.wikipedia.org/wiki/Interrupt_handler) (https://en.wikipedia.org/wiki/Interrupt_handler) execution times
- [Hard page faults](https://en.wikipedia.org/wiki/Page_fault) (https://en.wikipedia.org/wiki/Page_fault), see here for a [technical explanation](https://www.resplendence.com/whysoslow_help_hardpagefaults) (https://www.resplendence.com/whysoslow_help_hardpagefaults)

In other words it checks driver related problems & [more](https://www.resplendence.com/latencymon_idlt) (https://www.resplendence.com/latencymon_idlt) because the above mentioned checks are in direct relationship to it. LatencyMon itself does not solve any of the problems, what it does is to show possible root causes of the real latency problems under Windows.

Examples which caused a higher latency because of faulty drivers:

- [Samsung NVME 2.1 driver and terrible DPC Latency](https://hardforum.com/threads/samsung-nvme-2-1-driver-and-terrible-dpc-latency.1927697/) (https://hardforum.com/threads/samsung-nvme-2-1-driver-and-terrible-dpc-latency.1927697/)
- [Audio crackling due to nvlddmkm.sys Nvidia driver latency issue](https://www.tenforums.com/graphic-cards/26861-audio-crackling-due-to-nvlddmkm-sys-nvidia-driver-latency-issue.html) (https://www.tenforums.com/graphic-cards/26861-audio-crackling-due-to-nvlddmkm-sys-nvidia-driver-latency-issue.html)
- [In Depth latency test ryzen latency\(help\)](https://forums.guru3d.com/threads/in-depth-latency-test-ryzen-latency-help.422352/) (https://forums.guru3d.com/threads/in-depth-latency-test-ryzen-latency-help.422352/)

h3 Hard page faults

A [hard fault](https://appuals.com/hard-faults-per-second/) (https://appuals.com/hard-faults-per-second/) occurs when a memory block had to be retrieved from the Page File (Virtual Memory) instead of the physical memory (RAM). You have three possibilities, try to [disable the pagefile](https://tunecomp.net/win10-page-file-disable/) (https://tunecomp.net/win10-page-file-disable/) or upgrade your RAM (if possible).

h3 nVidia and dithering question

[Enabling dithering has no impact on gaming performance](https://www.youtube.com/watch?v=ot3TWFtWl1M) (https://www.youtube.com/watch?v=ot3TWFtWl1M), so it is unclear why nVidia and Microsoft are not going to enable the option (by default) for e.g. 1440p 144-165Hz G-Sync TN monitors (the ones which are most affected by color banding). Or why there is no option to manually control it via NVCP. AMD GPU drivers have such an option to enable/disable dithering since many years and Linux also has an active nVidia dithering option since many years. There is a [registry hack](https://forums.geforce.com/default/topic/1082681/geforce-drivers/is-it-possible-to-quot-port-quot-dithering-from-nvidia-x-server-to-geforce-driver-/post/5934577/#5934577) (https://forums.geforce.com/default/topic/1082681/geforce-drivers/is-it-possible-to-quot-port-quot-dithering-from-nvidia-x-server-to-geforce-driver-/post/5934577/#5934577) to enable dithering for nVidia users, the problem is that Windows seems to make problems (newer versions like 1903+).

h3 ULMB (nVidia) under Windows 10

ULMB is automatically grayed out because you can only use G-Sync OR ULMB at the same time. This is by design.

Most games under Windows 10 use the desktop refresh rate (default). Setting another refresh rate in your nVidia Control Panel might get ignored by the game itself (depending on the game). You can avoid this by disabling "Full-screen Optimisations", (if present - depends on the Windows 10 Build). Right-click on the game executable and check the compatibility tab, there is an option to enable it (it's by default disabled). After you enabled the option you **must** also enable the "Exclusive Fullscreen" option in-game.

h3 nVidia HDMI Audio problems with (wrong timeouts)

- **Warning:** You do NOT need this "trick" if you haven't installed NVIDIA High Definition Audio driver!

First of all, make a registry backup! There are several registry entries regarding on how to manipulate the timeout for HDMI Audio devices. Changing the value offsets *can fix AV Receiver connection* problems among an well known *audio stuttering* issue, this is because of "wrong" (unoptimized) timeouts. There is no easy way for this avbl. because those offsets change whenever you re-install or install a new nVidia/Realtek audio driver. What this registry change basically does is to force the device to go into a specific mode whenever the device is in a performance, idle or waiting stage.

- **Checking the Class GUID:** Open the device manager, navigate to "Sound, video and game controllers" and open the nVidia High Definition Audio properties. Now click the "Details" tab which reveals a bunch of GUIDs, we are searching for the "Class GUID", once you found it in the list, write down or right-click and copy it's value. The driver itself is called or listed as `nvhda64v.sys`.
- The value always looks different, the format is `{xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx}`.
- **Changing the value:** Open registry and navigate to `Computer\HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Control\Class\<-your-class-GUID-belongs-in-here>\0000\PowerSettings`. Once we are there, you see several entries, our focus are `ConservationIdleTime`, `IdlePowerState` & `PerformanceIdleTime`. The default value for all three of them is always (by default) `00 00 00 80`. If that's not the case you have to delete their content and ensure it's all set to zero -> `00 00 00 00`. In case you use a Realtek sound-chip, check sub-keys for `Realtek Semiconductor Corp.` and proceed with the same procedure.

- Reboot.

h3 Devices (common sense tips)

- Disable all integrated audio devices (e.g. nVidia/AMD audio drivers) e.g. if you use an external or add-on soundcard make sure you disable the internal sound chip unless you're a an *Audio Pro* which needs separate devices.
- Disable all unused integrated peripherals (like e.g. Network, Serial, Parallel Ports, etc.).
- Disable all unneeded running services (<http://www.blackviper.com/service-configurations/black-vipers-windows-10-service-configurations/>) via services.msc.

The more drivers Windows has to load the higher the chance is that it affects your IRQ/Latency/Processor. If you e.g. don't use nVidia HD audio driver(s) then just disable/uninstall or prevent from installing it in the first place. Windows then doesn't have to process separate devices or do additional calculations which results in a "smoother" gameplay (*lower latency ?*) experience.

h3 RAM

Windows 10 manages memory more efficiently than earlier versions of the OS, but more memory always can potentially speed up PC operations, games don't have to write/cache on the slower M2/SSD/HDD and can use the ram more efficient without the need to swap. *16 GB is the current default.*

Do not use e.g. 12 GB kit combinations because this will result in worse performance, as Intel explained [here](https://www.intel.com/content/www/us/en/support/articles/000005657/boards-and-kits.html#flex) (<https://www.intel.com/content/www/us/en/support/articles/000005657/boards-and-kits.html#flex>) the OS then falls back to a mode which is called "Flex mode".

h3 Interrupts

[Doctorzeus](https://forums.guru3d.com/members/doctorzeus.275790/) (<https://forums.guru3d.com/members/doctorzeus.275790/>) wrote a small utility called [MSIInterruptEnabler](https://github.com/TechtonicSoftware/MSIInterruptEnabler) (<https://github.com/TechtonicSoftware/MSIInterruptEnabler>) to enable Message Signal Interrupts on GPU's under Windows. It requires admin rights to use it and a reboot after you applied the new changes.

h3 Performance Monitor (perfmon.exe) + Windows Memory Diagnostic (mdsched.exe)

Windows has an integrated Performance Monitor program which checks (together with Eventmgr.msc) your OS/Hardware for problems, you should not disable it - since there are no noticeable performance improvements, keep an eye on it here and then to quickly check if there are failures or other problems. It is not necessary to install third-party tools such as HWInfo or AIDA (but in can help in case you quickly want to get a *better* overview).

h3 AMD Threadripper/Epyc and the Core Prio "Bug"

Indigo Bedroom Scene >	Linux	Windows
Threadripper 2990WX NUMA, Stock+XMP	~3.5	~1.5
Threadripper 2990WX, NUMA, No SMT	~2.9	~1.0
Epyc 7551, NUMA	~3.0	~1.3
Epyc 7551, UMA	~3.0	~3.0
Threadripper 2990WX, NUMA, Indigo in WSL	~1.55	
Threadripper 2950X, NUMA, Stock+XMP	~2.5	~2.3
Threadripper 2990WX 7zip, Compression	~90,000 MIPS	~41,000 MIPS
Threadripper 2990WX 7zip, Compression (with Coreprio fix from this article)	-	~70,000 MIPS

~~Threadripper and Epyc CPU's have a serious bug which crippled the performance, [Microsoft and AMD working together to fix this](https://community.amd.com/community/gaming/blog/2018/10/05/previewing-dynamic-local-mode-for-the-amd-ryzen-threadripper-wx-series-processors) (<https://community.amd.com/community/gaming/blog/2018/10/05/previewing-dynamic-local-mode-for-the-amd-ryzen-threadripper-wx-series-processors>), however as a workaround you can use [Bitsum's Coreprio](https://bitsum.com/portfolio/coreprio/) (<https://bitsum.com/portfolio/coreprio/>).~~

Fixed with Ryzen Master v2.0.1.1223 and with Windows 1909. Make sure you installed the latest BIOS and the latest Windows 10 patches. Ryzen Master will then **NOT SHOW THE Dynamic Local Mode** because this is not internally handled (fixed) by Windows. Coreprio is **NOT** needed anymore!

h3 Intel Transactional Synchronization Extensions (TSX)

[Intel Transactional Synchronization Extensions \(TSX\)](https://github.com/CHEF-KOCH/regtweaks/commit/6aef64ec092a3ab1704fdaf6c5b343aa44d2f1cf) (<https://github.com/CHEF-KOCH/regtweaks/commit/6aef64ec092a3ab1704fdaf6c5b343aa44d2f1cf>) is problematic, security and performance wise. Disabling Intel's TSX (closes Zombieload v2) on Haswell, Broadwell and

Skylake CPU's but [will affect the application as well as gaming performance](https://www.phoronix.com/scan.php?page=article&item=intel-jcc-gaming&num=2) (https://www.phoronix.com/scan.php?page=article&item=intel-jcc-gaming&num=2) around 4 up to 8%.

h3 MDS: Microarchitectural Data Sampling

Zombieload/MDS security holes affecting newer Intel CPU's (10th gen. except Icelake) as well as older architectures, the full list of affected CPU's are listed over [here](https://www.intel.com/content/www/us/en/security-center/advisory/intel-sa-00270.html) (https://www.intel.com/content/www/us/en/security-center/advisory/intel-sa-00270.html). You can quickly check if you're affected by the security hole with the official [MDS check tool](https://mdsattacks.com/files/mdstool-win-20190519.zip) (https://mdsattacks.com/files/mdstool-win-20190519.zip) (or via the script provided over <https://mdsattacks.com/> (https://mdsattacks.com/#ridl-ng), the tool/script checks the Intel CPU against Meltdown/MDS and other (current) open security holes.

h3 Lags

There are several kinds of [gaming releated lags](https://en.wikipedia.org/wiki/Input_lag) (https://en.wikipedia.org/wiki/Input_lag):

- CPU based lags - This causes the game to stutter, whenever the CPU reaches is max capabilities.
- Rendering based lags - This basically means your CPU & or GPU is to slow to handle the game/stream.
- General Network lag - This basically means your drivers or network are outdated/too slow or there are other interferences.
- [Stream Buffering](https://obsproject.com/wiki/Stream-Buffering-Issues) (https://obsproject.com/wiki/Stream-Buffering-Issues) - This basically means you use too much bitrate and your server which you are streaming at can't handle it.
- [Dropped Frames And General Connection Issues](https://obsproject.com/wiki/Dropped-Frames-and-General-Connection-Issues) (https://obsproject.com/wiki/Dropped-Frames-and-General-Connection-Issues) - This can have multiple sources, your ISP, your Router (or it's config), the streaming server or your AV program might blocks or intercept into the connection.
- Software based lags e.g. [Windows 10 Gaming Mode](https://obsproject.com/wiki/How-to-disable-Windows-10-Gaming-Features) (https://obsproject.com/wiki/How-to-disable-Windows-10-Gaming-Features) - Some Windows or OS related function can cause conflicts or making the game *hang*, this is mostly because of deadlocks (something hooks/inject specific code into another progress). Disabling Windows Game Mode, MSI Afterburner etc might already helps here. Another thing might be outdated software, it's true that an updated software/game/drivers can solves several performance issue, always keep an eye on the changelogs (release notes).

h3 Multi-GPU

Stay away from SLI/Crossfire it's [not worth](https://www.computerbase.de/2019-03/titan-rtx-sli-test/) (https://www.computerbase.de/2019-03/titan-rtx-sli-test/). There are too many unsolved problems since years and I consider it as "dead". To name several problems: Frame times, energy waste, unoptimized games & drivers. It's true that the FPS are higher on a multi-gpu system, however at what costs? Gaming on a single GPU is in most cases good enough unless you play on super high resolutions (5k+).

h3 Nvidia FPS Limiter v3 versus RTSS

Starting with [441.87](https://github.com/CHEF-KOCH/nVidia-modded-Inf/releases/tag/441.87) (https://github.com/CHEF-KOCH/nVidia-modded-Inf/releases/tag/441.87) nVidia added their own FPS Limiter (v3) in the NV control panel. You might [ask yourself](https://youtu.be/W66pTe8YM2s) (https://youtu.be/W66pTe8YM2s) what are the differences between [RTSS](https://www.guru3d.com/files-details/rtss-rivatuner-statistics-server-download.html) (https://www.guru3d.com/files-details/rtss-rivatuner-statistics-server-download.html) and nVidia's solution and which solution should you prefer?

Overall:

- The frametime is more consistent with RTSS because this is how RTSS works.
- nVidia's solution does not allow you to change the frame rate without restarting the game, this works in RTSS. nVidia could fix or change that in the future.
- Every overlay you use, no matter what program will add a small delay!
- NVIDIA Inspector old limiter (v2) and AMDs Chill adding the most input delay.

RTSS Advantages:

- Nice graphical user interface
- RTSS settings are easy to back up
- Decimal frame limits can be set (to reduce V-Sync latency), which other tools cannot do
- Allows per-game custom settings
- RTSS supports both global settings and individual executable settings

Cons:

- Adds a frame of [input delay](https://www.youtube.com/watch?v=rsOPYCpBJjc) (https://www.youtube.com/watch?v=rsOPYCpBJjc), to workaround it use the default ("Global" profile)
- It's an separate third-party tool which needs to run in the background
- Can interfere with steam's overlay feature
- Can crash specific games, depending on the settings and used injection method
- Some games will not start while injecting the overlay into it
- Some games can detect the injection method as "cheat" and disconnect or at worst, ban you!

nVidia Frame Limiter Advantages:

- No third-party software required! Minimal background services running.
- No per-game setting, you need to set the frame limit globally OR via the specific game profile which is a bit clunky

Cons:

- No *fancy* GUI
- Not as advanced as RTSS
- Some games (once the limiter is enabled) having problems showing the in-game overlays

h3 FreeSync Gaming on Nvidia (Adaptive Sync)

Open the Nvidia Control Panel, browse to “Set up G-Sync”, then select your FreeSync monitor. From here, make sure both the “enable G-Sync, G-Sync compatible” checkbox and the “enable settings for the selected display model” checkbox are ticked. The second checkbox doesn’t appear if the monitor is G-Sync certified. Then click Apply, your monitor will restart and adaptive sync will be enabled.

h3 Control Panel and it's toggles

TweakGuides has a full documentation what each toggle/settings in the nVidia Control Panel (NVCP) does, I highly suggest that you read it before you touch any of the settings. The Guide constantly gets updates as soon nVidia changes something in the NVCP. The default settings are usually fine and you might only want to tweak the application/game settings related profiles, which you can do via the NVCP or [nVidia Inspector](https://github.com/DeadManWalkingTO/NVidiaProfileInspectorDmW) (https://github.com/DeadManWalkingTO/NVidiaProfileInspectorDmW) (unofficial utility).

h3 nVidia Shader Cache on a HDD

Shaders are loaded by the game itself, like any other of their assets, and sent to the driver for compilation (meaning transformed from source code form into binary form that can be used by the GPU.) The driver compiles them and sends the results back to the game. The compilation happens by the CPU, not the GPU, and is slow. A shader cache doesn't get rid of the loading-from-disk step. It only replaces the compilation step with a second loading step (loading the previously compiled form of the shader from disk and sending that to the game.) Even with a very slow disk, loading small already compiled shader files from said disk is much faster than generating them again. The shader cache is meant to persist between reboots. Putting them on a RAM disk meaning losing the cache on reboot - unless you're syncing the RAM disk to disk on reboot.

h3 How to get Windows 10 custom resolutions in any aspect ratio you want

[jim2point0](https://www.deadendthrills.com/forum/discussion/504/how-to-custom-aspect-ratios-and-resolutions-via-dsr-nvidia-only) (https://www.deadendthrills.com/forum/discussion/504/how-to-custom-aspect-ratios-and-resolutions-via-dsr-nvidia-only) wrote an excellent guide how to *optimize* nVidia's DSR via registry in order to get any resolution you want.

h3 Texture Filtering - Negative LOD Bias (LOD-BIAS) - Is it cheating or not?

On nVidia graphic cards (GPU's) you can use nVidia Inspector in order to change the [LOD-BIAS offset which make you game look like a cartoon](https://www.hardocp.com/news/2019/01/21/does_adjusting_graphics_settings_lower_than_game_allows_constitute_cheating63/) (https://www.hardocp.com/news/2019/01/21/does_adjusting_graphics_settings_lower_than_game_allows_constitute_cheating63/) while I would argue that this has to be patched out of the game (because this can't be used by AMD users and gives you a unfair benefit) I overall say that this doesn't deserves to be banned.

Keep in mind that this isn't allowed in eSports and even a banned topic/"trick" on a normal casual game like Battlefield 5, Overwatch and Quake Champions.

However, it's true that his gives you a massive FPS boost.

h2 Patch Related Performance Impact

h3 KB4482887

[KB4482887](https://support.microsoft.com/en-au/help/4482887/windows-10-update-kb4482887) (https://support.microsoft.com/en-au/help/4482887/windows-10-update-kb4482887) does *for some users* decrease the gaming performance, however it does [not have anything to do with Retpoline](https://old.reddit.com/r/microsoft/comments/ax18s7/kb4482887_caused_major_performance_issues_for_me/ehrrtr4b/) (https://old.reddit.com/r/microsoft/comments/ax18s7/kb4482887_caused_major_performance_issues_for_me/ehrrtr4b/) changes because it's [not enabled by default](https://www.borncity.com/blog/2019/03/05/windows-10-retpoline-spectre-2-schutz-manuell-aktivieren/) (https://www.borncity.com/blog/2019/03/05/windows-10-retpoline-spectre-2-schutz-manuell-aktivieren/). Microsoft official says that you can ignore/uninstall or [hide](https://support.microsoft.com/de-de/help/4026726/windows-hide-windows-updates-or-driver-updates) (https://support.microsoft.com/de-de/help/4026726/windows-hide-windows-updates-or-driver-updates) the update. They promise to fix the performance issue with the next patchday.

// Microsoft statement
After installing KB4482887, users may notice graphics and mouse performance degradation with desktop gaming when playing certain games (eg: Destiny 2)"...

Fixed with Windows 10 Build 17763.379 and [KB4489899](https://support.microsoft.com/en-us/help/4489899/windows-10-update-kb4489899) (https://support.microsoft.com/en-us/help/4489899/windows-10-update-kb4489899).

h2 Build 1809 17763.503 - KB4494441

[Retpoline](https://techcommunity.microsoft.com/t5/Windows-Kernel-Internals/Mitigating-Spectre-variant-2-with-Retpoline-on-Windows/ba-p/295618) (https://techcommunity.microsoft.com/t5/Windows-Kernel-Internals/Mitigating-Spectre-variant-2-with-Retpoline-on-Windows/ba-p/295618) is activated by default with [KB4494441](https://support.microsoft.com/en-us/help/4494441/windows-10-update-kb4494441) (https://support.microsoft.com/en-us/help/4494441/windows-10-update-kb4494441) if Spectre Variante 2 (CVE-2017-5715) was activated (manually) before the KB was installed.

h2 Build 1903 - KB4515384 (Audio Gaming Issue)

This KB introduced several Audio API changes, as a result people reported that their in-game sound was very low, muffled or "strange". Some people even said that this might causes several games to "hang" (Overwatch, Apex Legends, WoW Classic, and PlayerUnknown's Battlegrounds (PUBG)) for a short amount of time.

Microsoft seems aware of the issue and provides a workaround.

- "To mitigate the issue, open settings in the impacted game and disable multi-channel audio, if this option is available. You can also search in the Windows Control Panel for third-party audio device control panels and disable Multi-channel audio or Virtual Surround Sound, if these options are available.
- We are working on a resolution and estimate a solution will be available in late September."

Another workaround is, in case a game does not provides such an option, to change your audio properties within your sound cards control panel. It's not recommend to uninstall the update because it fixes over 79 vulnerabilities, it also fixes a SearchUI.exe CPU usage problem.

h2 Monitor overclocking

- [GamersNexus](https://www.gamernexus.net/guides/1674-overclock-monitor-higher-refresh-rate) (https://www.gamernexus.net/guides/1674-overclock-monitor-higher-refresh-rate) has a nice guide to get started into the topic - worth to read!

Why overclock your Monitor?

- Higher Refresh Rate(s)
- Custom Resolution
- Fix/Correct missing/wrong [EDID](https://en.wikipedia.org/wiki/Extended_Display_Identification_Data) (https://en.wikipedia.org/wiki/Extended_Display_Identification_Data) values
- Correct/change or expand FreeSync range(s)
- Smoother fps gameplay*

Possible Disadvantages

- Artifacts or no picture output
- The settings you made must be imported and applied manually after each new driver installation (import/export)
- Skipped frames (even if the monitor accepted the new values and everything seems "fluid")
- HDCP might break or DRM fails
- Color *might be* changed after applying the changes use [Color Sustainer](http://www.guru3d.com/files-details/color-sustainer-download.html) (http://www.guru3d.com/files-details/color-sustainer-download.html) to correct this. (This is a OS issue!)

Possible Risks?

None (!), every change is temporarily until you approved the change. You can import/export profiles and backup the current states before you do any serious change. The hardware itself will not be affected in case something went wrong.

Tools needed

- [Custom Resolution Utility \(CRU\)](https://www.monitortests.com/forum/Thread-Custom-Resolution-Utility-CRU) (https://www.monitortests.com/forum/Thread-Custom-Resolution-Utility-CRU)
- (optional) [AMD/ATI Pixel Clock Patcher](https://www.monitortests.com/forum/Thread-AMD-ATI-Pixel-Clock-Patcher) (https://www.monitortests.com/forum/Thread-AMD-ATI-Pixel-Clock-Patcher)
- (optional) [NVIDIA Pixel Clock Patcher](https://www.monitortests.com/forum/Thread-NVIDIA-Pixel-Clock-Patcher) (https://www.monitortests.com/forum/Thread-NVIDIA-Pixel-Clock-Patcher)

Make sure you install a correct [ICC Color profile](http://www.tftcentral.co.uk/articles/icc_profiles.htm#install) (http://www.tftcentral.co.uk/articles/icc_profiles.htm#install). You can right-click on it and install it or place it manually under `\System32\Spool\Drivers\Color`. You should work with two profiles, one for applications such as Office and one for Gaming (if you have a separate monitor for *gaming only* you can skip this part and only apply the gaming color profile).

If possible:

- Go with an IPS panel (yep, even for gaming such as the ASUS ROG PG279Q)
- See if you get a compromise between HZ, Color and viewing angle
- Good reviews are coming from professional sites like [prad.de](https://www.prad.de/) (https://www.prad.de/), [TFT Central](http://www.tftcentral.co.uk/) (http://www.tftcentral.co.uk/) or [ExpertReviews](https://www.expertreviews.co.uk/accessories/pc-monitors) (https://www.expertreviews.co.uk/accessories/pc-monitors).
- Make sure that you correctly [calibrate your Monitor](https://www.digitaltrends.com/computing/how-to-calibrate-your-monitor/) (https://www.digitaltrends.com/computing/how-to-calibrate-your-monitor/). In case that you can effort a hardware Colorimeter such as the [i1Display Pro](https://www.xrite.com/categories/calibration-profiling/i1display-pro) (https://www.xrite.com/categories/calibration-profiling/i1display-pro) prefer this instead.

h2 FPS frame rate limiter

Why is capping the FPS important?

There is an detailed answer [here](https://www.blurbusters.com/howto-low-lag-vsync-on/) (https://www.blurbusters.com/howto-low-lag-vsync-on/) in short: you should **ALWAYS CAP THE FPS** which results in a smooth stutter and lag-free experience. RTSS Rivatuner Statistics Server is more accurate than most (not all) in-game FPS limiter, the nVidia own limiter adds some delay to it and is not recommend to use!

h2 Multi-monitor performance issues

Multiple monitors plugged *might* causes performance issues which is depending on the game engine and if you do other tasks to the same time on other monitors. A workaround is to Unplug non-primary monitors or to ensure that every monitor runs at the same Hz.

h2 Enable Variable Refresh Rate (FreeSync)

The new [VRR](https://devblogs.microsoft.com/directx/os-variable-refresh-rate/) (https://devblogs.microsoft.com/directx/os-variable-refresh-rate/) **does not** replace the traditional FreeSync / Adaptive-Sync! The new Windows toggle doesn't override any of the settings you've already configured in the FreeSync or Adaptive-Sync control panels.

Requirements:

- Windows 1903+
- WDDM 2.6+ driver
- FreeSync or Adaptive-Sync capable monitor
- DX11 full-screen game which does not support VRR natively

h2 Enabling dithering for nVidia Geforce GPUs (non-Quadro GPUs/Linux)

Worth knowing

- Under AMD Temporal dithering is (by default) on.
- [Intel Graphics tweak tool](https://github.com/nitrocaster/igfxtweak) (https://github.com/nitrocaster/igfxtweak) can adjust Intel's dithering. Newer Intel driver have dithering disabled by default.
- Programs like Handbreak automatically seems to apply dither when converting to 8 bit.
- Since nVidia driver v353.06 users can select bit depth in the Nvidia Control Panel on all systems, if you see 10-bit Nvidia will automatically dither down to the bit depth selected if it is lower. If you can select 10-bit in Nvidia CP, the driver will not dither and sent true 10bit signal to your display.
- Nvidia will not dither 10-bit input when set to 12-bit.

Dithering on Nvidia GPUs are usually only present with Quadro GPUs, the only [exception are GeForce drivers under any Linux OS](https://forums.geforce.com/default/topic/1082681/geforce-drivers/is-it-possible-to-quot-port-quot-dithering-from-nvidia-x-server-to-geforce-driver-/post/5934577/#5934577) (https://us.download.nvidia.com/XFree86/Linux-x86/375.10/README/xconfigoptions.html). The user [Guzz](https://forums.geforce.com/member/1822975/) (https://forums.geforce.com/member/1822975/) discovered a way to [unlock the hidden registry entry "DitherRegistryKey"](https://forums.geforce.com/default/topic/1082681/geforce-drivers/is-it-possible-to-quot-port-quot-dithering-from-nvidia-x-server-to-geforce-driver-/post/5934577/#5934577) (https://forums.geforce.com/default/topic/1082681/geforce-drivers/is-it-possible-to-quot-port-quot-dithering-from-nvidia-x-server-to-geforce-driver-/post/5934577/#5934577). **The method seems to be stable for Windows 7 up to Win 10 1607. Later OS versions seems to have massive problems. Linux handles it a bit different from Windows** (https://help.teradici.com/s/article/1048).

Drawbacks

- Constant [banding](https://loopit.dk/banding_in_games.pdf) (https://loopit.dk/banding_in_games.pdf) everywhere on 1903+ (?)
- Disable dithering -> shut down -> turn on PC -> enable dithering seems the best way to workaround the problems.

- Nvidia drivers (300+) rounding VCGT from 16-bit to 8-bit with certain scenario such as after monitor wake up from sleep or after computer wake up from sleep which might causes problems.
- Multi-Monitor problems.

h2 What is mouse smoothing and why is it bad for gaming?

Modern gaming [mice](https://en.wikipedia.org/wiki/Computer_mouse) (https://en.wikipedia.org/wiki/Computer_mouse) are *very accurate* when it comes to reporting where their users are moving the mouse, but older mouse sensors were not nearly as accurate as the mice with optical sensors that we have nowadays.

Overall spoken, [mouse smoothing](https://www.codeproject.com/Articles/5247322/Mouse-Smoothing) (<https://www.codeproject.com/Articles/5247322/Mouse-Smoothing>) is the interpolation of raw input data from the mouse in order to prevent jerky movement in low resolution mice while Mouse acceleration is a setting where the cursor distances increases if the mouse is moved quickly. While the setting is a **personal choice**, it is recommend that you disable mouse acceleration to provide a more consistent aim in more competitive titles.

Gaming wise you want your mouse to register your movements as "raw" as possible. Any software or hardware based attempt to alter what is being shown on screen can negatively affect your [input lag performance](https://old.reddit.com/r/MouseReview/comments/eri3cx/some_fps_games_feel_more_responsive_than_others/) (https://old.reddit.com/r/MouseReview/comments/eri3cx/some_fps_games_feel_more_responsive_than_others/).

[RInput](https://fearless-assassins.com/files/file/1656-rinput/) (<https://fearless-assassins.com/files/file/1656-rinput/>) is probably the best tool used to remove mouse acceleration from GoldSrc games. This tool does NOT remove mouse acceleration from your system, the OS itself has an option for it.

Mouse tweaks overall provides more detailed configuration of mouse cursor behavior. It provides a range of accessibility enhancements as well a power-user features, including mouse gestures.

h3 How to tell if mouse acceleration is enabled

Mouse acceleration increases the speed of your mouse cursor based on the speed you move your mouse. This might sound like a good idea but it causes problems - if you move your mouse from point A to point B then you'd expect your cursor to do the same thing each time you move the mouse between those two points; with acceleration enabled it won't because you'll vary the speed that you move the mouse between those two points and end up with inconsistent aiming.

You can do this within Windows 10 "Mouse settings" or via registry. Some engines (games) providing also an option via GUI/CMD to adjust the "Raw Input", check if you game includes such an option before you globally disable pointer precision.

In some situations Windows 10 or your mice driver do not show any option to disable it, so the most reliable solution is to toggles this via registry. The corresponding key is located at `HKEY_CURRENT_USER\Control Panel\Mouse` , we want to change the two entries called `SmoothMouseXCurve` and `SmoothMouseYCurve` .



```
//SmoothMouseXCurve

00,00,00,00,00,00,00,00

00,a0,00,00,00,00,00,00

00,40,01,00,00,00,00,00

00,80,02,00,00,00,00,00

00,00,05,00,00,00,00,00

//SmoothMouseYCurve

00,00,00,00,00,00,00,00

66,a6,02,00,00,00,00,00

cd,4c,05,00,00,00,00,00

a0,99,0a,00,00,00,00,00

38,33,15,00,00,00,00,00
```

You do not need any other "input lag mouse fixes"!

h3 How to remove mouse acceleration?

There are two basic types of mouse acceleration; "positive" acceleration and "negative" acceleration. Both are interpolation rates. Positive means that the faster you are moving your mouse, the farther cursor will travel on the screen. Negative means that the faster your mouse is moving, the less distance cursor will be travel on screen.

- Launch your fav. game and go into a corner with no enemies or something which could desturb you.
- Move your crosshair to a specific spot, for example the edge of a building, distant window etc. This basically will act like your starting point from were you begin. Another good example is a wall with some tiles.
- Put your hand on the mousepad.
- Move the mouse slowly from your hand and create 90° / 180° rotation in-game. This movement should take > 2 seconds.
- Now move the mouse back to the default position marked by your hand as fast as you can.
- If your crosshair ended up roughly on the same spot, the game does not feature acceleration. Usually when games use acceleration, your crosshair will end up on a vastly different location.

Another method would be to open the debug console (if possible) and check if a mouse related acceleration or interpolation entry exists which you then could adjust.

h3 Mouse polling rate

- You could change your [mouse polling rate](https://www.howtogeek.com/182702/mouse-dpi-and-polling-rates-explained-do-they-matter-for-gaming/) (https://www.howtogeek.com/182702/mouse-dpi-and-polling-rates-explained-do-they-matter-for-gaming/) down from 1000hz to 250hz/120hz because *sometimes* it can reduce the overall mouse lag for some specific games or game engines. Keep in mind that this tweak is *optional* because in most of the cases 1000hz works just fine (depending on the used mouse driver).

h3 Sensitivity

Take your [mouse sensitivity from another game](https://www.mouse-sensitivity.com/) (https://www.mouse-sensitivity.com/), see if it helps you but at the end you must decide if you like it or not.

h3 Placebo "tweaks" and promises

There is no 'pro aim' tweak, just give yourself time to progress. Play hard go pro - yep it's true!

h1 Software and drivers you don't need.

h3 Windows

- [ReyonMouseFix](https://github.com/e9k/ReyonMouseFix) (https://github.com/e9k/ReyonMouseFix)
- [Povohat Mouse Accel / Interaccel](https://mouseaccel.blogspot.com/2015/12/new-method-for-mouse-acceleration.html) (https://mouseaccel.blogspot.com/2015/12/new-method-for-mouse-acceleration.html)
- [The MarkC Windows 10 + 8.1 + 8 + 7 Mouse Acceleration Fix](https://donewmouseaccel.blogspot.com/2010/03/markc-windows-7-mouse-acceleration-fix.html) (https://donewmouseaccel.blogspot.com/2010/03/markc-windows-7-mouse-acceleration-fix.html)

h3 Linux

- [A modded version of the Linux USB mouse driver with acceleration](https://github.com/chilliams/mousedriver) (https://github.com/chilliams/mousedriver)

h3 Can I get banned for using a modded mice driver?

No.

h3 Why you shouldn't use modified drivers?

I don't recommend it e.g. povohat accel to [new players](https://docs.google.com/spreadsheets/d/1UaM765-S515ibLyPaBtMnBz7xiao0HL5f-F1zk_CSf4/edit#gid=1901373467) (https://docs.google.com/spreadsheets/d/1UaM765-S515ibLyPaBtMnBz7xiao0HL5f-F1zk_CSf4/edit#gid=1901373467), because it's better to understand what sensitive ranges YOU like before adding accel in there. Most users probably wouldn't know what settings to use.

Key point:

- Windows 10 got better
- Mouse vendors improved their drivers and mice software
- Windows 8 mouse accel problems got fixed in Windows 10
- *Performance issue* (in theory), above mentioned drivers DO NOT causing performance issue but there others (even older drivers) who are known to be problematic

Downsides with modded drivers

- Other software might not work e.g. FaceIT.
- Unsigned drivers which possesses a security risk, you need to disable Windows 10 driver signing to load unknown drivers from "unknown" places.

- LED and some DPI features might not work.
- You MUST uninstall all current mice drivers BEFORE you install new drivers.
- Uninstall process might not be easy (for beginners) since most drivers or tweaked software mice drivers lack of a proper uninstaller (leftover files).

h3 What to do instead?

- Just enable raw input in-game. When it's enabled, the game takes input straight from the mouse and not from the operating system.
- Disabling enhanced pointer precision is recommended but [not necessary](https://youtu.be/F2-_hkv3gDA) (https://youtu.be/F2-_hkv3gDA).
- Ensure you use latest mice driver for your mice. If you find problems, report them to the developers, so that they fix it.
- If you game supports "fine tweaking" mice settings via e.g. m_customaccel or/and m_customaccel_exponent you can adjust it as per own needs. Keep in mind not all games allowing you to tweak those settings.

h3 Disable Windows 10 driver updates delivered through Windows Update

See ["How to temporarily prevent a driver update from reinstalling in Windows 10"](https://support.microsoft.com/en-us/help/3073930/how-to-temporarily-prevent-a-driver-update-from-reinstalling-in-windows-10) (https://support.microsoft.com/en-us/help/3073930/how-to-temporarily-prevent-a-driver-update-from-reinstalling-in-windows-10). Another alternative is using [WuMGr](https://github.com/DavidXanatos/wumgr) (https://github.com/DavidXanatos/wumgr) which is basically a front-end for Windows crippled Windows Update interface, it basically spoken can take control over the Windows Updater mechanism by e.g. settings registry/gpo settings and also includes some more options to handle updates.

h3 Wi-Fi scanning can cause 'latency' spikes

Some [Intel wireless adapters](https://www.intel.com/content/www/us/en/support/articles/000005546/network-and-i-o/wireless-networking.html) (https://www.intel.com/content/www/us/en/support/articles/000005546/network-and-i-o/wireless-networking.html) and [Atheros adapters](https://superuser.com/questions/881880/turn-off-wi-fi-scanning-on-windows-8) (https://superuser.com/questions/881880/turn-off-wi-fi-scanning-on-windows-8) are known to cause small [latency spikes](https://blog.mojonetworks.com/background-scanning-is-dead) (https://blog.mojonetworks.com/background-scanning-is-dead) (*original link dead!*). On older operating systems, [WLAN Optimizer](http://www.martin-majowski.de/) (http://www.martin-majowski.de/) was a workaround for these issues. Since [Windows 10](https://answers.microsoft.com/en-us/windows/forum/windows_10-networking/is-there-any-way-to-stop-windows-10-from-scanning/3870b3d1-0f07-4875-8779-bb5c11fce0a8) (https://answers.microsoft.com/en-us/windows/forum/windows_10-networking/is-there-any-way-to-stop-windows-10-from-scanning/3870b3d1-0f07-4875-8779-bb5c11fce0a8), you can do this directly in Windows [without any tool](https://pastebin.com/WAdTPBpC) (https://pastebin.com/WAdTPBpC).

Some newer drivers also include an option in the network adapter settings to configure/enable/disable the Wi-Fi scan interval.

h5 Disable Wi-Fi background scanning for Atheros adapters

- Go to the **Control Panel** .
- Choose **Networks and Sharing** and **Change Adapter Settings** .
- Right-click on your **Wi-Fi adapter icon** and select **Properties** .
- Click **Configure** which is located under **Connect Using** .
- Go to the **Advanced** tab.
- Set **Background Scan** to **Disabled** .
- Set **Foreground Scan** to **Disabled** .

h5 Disable Wi-Fi background scanning for Atheros adapters

- Type **ncpa.cpl** into the Windows search box which opens the Network Connectivity Manager
- Right-click on the **Wi-Fi adapter icon** and select **Properties** .
- Click the **Configure** button on the driver property page.
- On the **Advanced tab** , scroll down the **Property** list and set **Roaming Aggressiveness** to **Lowest** .
- Click **OK** to apply it to all opened settings.

h5 Disable wireless autoconfig globally

Some Windows 10 builds (not all, some need to change the registry toggle because it's locked service task) allow you to disable the **Wireless Autoconfig service** (https://windowsreport.com/make-windows-stop-searching-wireless-networks/), which prevents Windows from searching for new Wi-Fi networks in general (*useful if you're only connected via ethernet*).

h3 Intel adapter 'recommended' network adapter settings

Network adapter settings are usually accessed through it's driver properties. From "Device Manager", or by right-clicking the adapter connection, choosing properties and then configure.

Keep in mind that some drivers might have different settings/options! After talking with NordVPN I figured out with them that some settings might directly have a negative effect on Down-/Upload speed. Ensure you test each settings one by one. In this case try to disable all "offloading" params.

h4 Fast response and low-latency optimization (gaming):

- Disable Interrupt Moderation Rate (if not possible decrease the size)

- Disable Offload TCP Segmentation
- Increase Transmit Descriptors
- Increase Receive Descriptors
- Increase RSS Queues

h4 Throughput optimization (e.g. servers):

- Enable Jumbo Frames
- Increase Transmit Descriptors
- Increase Receive Descriptors

h4 For low CPU usage:

- Maximize Interrupt Moderation Rate
- Keep Receive Descriptors at default
- Avoid setting large Receive Descriptors
- Decrease RSS Queues
- Decrease the Max number of RSS CPUs in Hyper-V environments

h4 Optional disable/change the following options on your network adapter

- **Disable** Adaptive Inter-Frame Spacing
- **Disable** Flow Control
- **Disable** Interrupt Moderation
- Set Interrupt Moderation Rate to **OFF** (not needed if you disable Interrupt Moderation)
- Set Enable PME to **Disabled**
- **Disable** all “Offload” features
- **Disable** Packet Priority & VLAN
- **Disable** Jumbo Packet
- Set Receive Side Scaling (RSS) - **ENABLED**
- RSS Balancing Mode - **NUMAScaling**
- **Do not touch Speed(?rate) & Duplex, leave it on auto-negotiation**

h4 Adjust via CMD/PowerShell

```
netsh interface teredo set state disabled
netsh interface 6to4 set state disabled
netsh winsock reset
netsh interface isatap set state disable
netsh int tcp set global timestamps=disabled
netsh int tcp set heuristics disabled
netsh int tcp set global autotuninglevel=disable
netsh int tcp set global congestionprovider=ctcp
netsh int tcp set supplemental Internet congestionprovider=CTCP
netsh int tcp set global chimney=disabled
netsh int tcp set global ecncapability=disabled
netsh int tcp set global rss=enabled
netsh int tcp set global rsc=disabled
netsh int tcp set global dca=enabled
netsh int tcp set global netdma=enabled
PowerShell Disable-NetAdapterChecksumOffload -Name "_ "
PowerShell Disable-NetAdapterLso -Name "_ "
PowerShell Disable-NetAdapterRsc -Name "_ "
PowerShell Disable-NetAdapterIPsecOffload -Name "_ "
PowerShell Disable-NetAdapterPowerManagement -Name "_ "
PowerShell Disable-NetAdapterQos -Name "_ "
```

h4 Enable or disable QoS?

It depends on how you set up your QoS/WMM. QoS is useful in scenarios where your network experiences peak or near-peak loads on a routine basis. If the network is not under significant load, QoS rules will not be applied, the same goes if your router is not configured to use QoS or the firmware doesn't includes it.

For example, if you live alone, don't play online games, don't use VOIP, don't use video streaming services (Netflix/Amazon/Hulu/Twitch.tv & YouTube), or don't use P2P file sharing - QoS is not going to be all that beneficial to you.

- Check if you're ISP offers a QoS package to residential connections (most do)
- Check your Router QoS settings
- Check your OS QoS options

In general **I recommend leaving it enabled**. If there some problems check your configuration instead, I did some benchmarks and I never found any impact (because my router isn't configured to use it). It's right that Windows reserves some bandwidth but that doesn't matter because your router might ignore the flags.

h3 Optional network tweaks

h2 Disable "Inbound Layer 7" on your router

Layer 7 filtering or shaping is identifying traffic at layer 7 (Application Layer) of the OSI model. Instead of shaping/filtering based on the port and source/destination, you are identifying a stream based on its contents. This is also sometimes called deep packet inspection since it works by looking into the contents of the packets not just the headers. If you're concerned about performance: IPP2P and especially L7 are slower than simple IP, MAC or port matches. L7 can't cope well with encrypted P2P traffic in background while gaming.

- Enable SYN cookies
- Disable IPv6 tunnel adapter & interfaces
- Disable all ISATAP, 6to4 and Teredo Tunneling interfaces
- Ensure Windows Defender Core Isolation stays disabled (default) it can causes 'lags' with several older games
- Disable the Timeline function, it eats HDD space in order to collect your history or if you use it, ensure e.g. CCleaner cleans it - > bigger file means longer loading times.

```
>-

netsh interface teredo set state disable
netsh interface 6to4 set state disable disable
netsh interface isatap set state disable
REG ADD "HKLM\SYSTEM\CurrentControlSet\services\TCPIP6\Parameters" /v "EnableICSIPv6" /t REG_DWORD /d 0 /f
REG ADD "HKLM\SYSTEM\CurrentControlSet\services\TCPIP6\Parameters" /v "DisabledComponents" /t REG_DWORD /d 255 /f
```

Keep in mind that every modern Router firmware has an option to block or at least filter Teredo traffic, same like NetBios which means you don't need to 'disable' everything blindly because nothing can pass the router anyway!

h2 Reducing the network latency to eliminate microstuttering in games

The [Multimedia Class Scheduler](https://docs.microsoft.com/en-us/windows/win32/procthread/multimedia-class-scheduler-service?redirectedfrom=MSDN) (https://docs.microsoft.com/en-us/windows/win32/procthread/multimedia-class-scheduler-service?redirectedfrom=MSDN) Service has built-in network throttling which can degrade performance of online games. That's the theory.

This [network tweak](https://www.speedguide.net/articles/gaming-tweaks-5812) (https://www.speedguide.net/articles/gaming-tweaks-5812) aims to reduce the network latency, however this tweak is [controversial](http://forum.sudden-strike-alliance.fr/all-problem-with-lag-on-windows-7-and-10-t2807.html) (http://forum.sudden-strike-alliance.fr/all-problem-with-lag-on-windows-7-and-10-t2807.html) because modern network cards/adapters already handlining it very well. The values are depending on `AvSetMmMaxThreadCharacteristics()` & `AvSetMmThreadCharacteristics()` and what Windows 10 version you use, most mentioned entries are already set by default to a good value.

Whenever you load a thread for this e.g. while gaming it must use these [two API's](https://www.installsetupconfig.com/win32programming/windowsthreadsprocessapis7_4.html) (https://www.installsetupconfig.com/win32programming/windowsthreadsprocessapis7_4.html) otherwise nothing will happen. So this depends how the game implemented the netcode. More information regarding this topic is avbl. over [here](http://msdn2.microsoft.com/en-us/library/ms681973.aspx) (http://msdn2.microsoft.com/en-us/library/ms681973.aspx), [here](http://msdn2.microsoft.com/en-us/library/ms681974.aspx) (http://msdn2.microsoft.com/en-us/library/ms681974.aspx) and [here](https://docs.microsoft.com/en-us/windows/win32/procthread/multimedia-class-scheduler-service?redirectedfrom=MSDN) (https://docs.microsoft.com/en-us/windows/win32/procthread/multimedia-class-scheduler-service?redirectedfrom=MSDN).

The task priority ranges from 1, which represents "low" to 8, which represent "high". However, the scheduling category overrides this, so that if scheduling category is automatically to "High", then the priority value is automatically set to 2 (unless gaming mode was activated and the game supports it).

```
>-

Windows Registry Editor Version 5.00

; I do not recommended using this, since changing the defaults
; Starting with Windows 1909 this is not anymore needed.
; https://github.com/CHEF-KOCH/GamingTweaks/issues/5

[HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Multimedia\SystemProfile]
```

"NetworkThrottlingIndex"=dword:ffffff
"SystemResponsiveness"=dword:00000000
[HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Multimedia\SystemProfile\Tasks\Games]
"Scheduling Category"="High"
"SFIO Priority"="High"
"Background Only"="False"
"Priority"=dword:00000001
"Clock Rate"=dword:00002710
"GPU Priority"=dword:00000001
"Affinity"=dword:00000000

[Multimedia streaming](http://msdn2.microsoft.com/en-us/library/ms684247.aspx) (<http://msdn2.microsoft.com/en-us/library/ms684247.aspx>) and some other games that uses [Multimedia Class Scheduler service \(MMCSS\)](https://docs.microsoft.com/en-us/windows/desktop/procthread/multimedia-class-scheduler-service) (<https://docs.microsoft.com/en-us/windows/desktop/procthread/multimedia-class-scheduler-service>) can only utilize up to 80% of your CPU. MMCSS ensures prioritized access to CPU resources, without denying CPU resources to lower-priority background applications.

verdict

If [the game](https://www.reddit.com/r/GlobalOffensive/comments/646v5q/fps_boost_with_windows_10_update/) (https://www.reddit.com/r/GlobalOffensive/comments/646v5q/fps_boost_with_windows_10_update/) explicitly supports game mode, then this old "tweak" is redundant (after Windows 10 1809+ due to Game Mode changes) (*assuming you use + enabled Game Mode*). If the netnode differs or was optimized then it's also obsolete since the game internally handles it differently. So overall, spoken it's depending on the game, one example is the old Call of Duty, this game can get a benefit from using the mentioned tweak but that's the only game I found (so far).

Windows 10 and SafeDisc

SafeDisc games do not work on Windows 10 and restoring the ability to load SafeDisc protected games will make your OS vulnerable to several attacks.

- Extract a temporary location on the system.
- Copy `secdrv.sys` to `%WINDIR%\system32\drivers` .
- Import `secdrv.reg` to Windows registry.
- As the service may still not work due to lack of a [digital signature](https://windowsreport.com/driver-signature-enforcement-windows-10/) (<https://windowsreport.com/driver-signature-enforcement-windows-10/>), you need to disable it.

MIDI Mapper

Microsoft officially removed the MIDI Mapper support in Windows 10. You can use [CoolSoft MIDIMapper instead](https://coolsoft.altervista.org/en/midimapper) (<https://coolsoft.altervista.org/en/midimapper>).

Alt F4 doesn't terminate frozen applications

This is not OS related, some games utilize the Windows manager differently, as a alternative you can use [Super F4](https://stefansundin.github.io/superf4/) (<https://stefansundin.github.io/superf4/>).

Why is Core Parking disabled in Windows 10?

Microsoft officially states that "[Core Parking is most effective on systems that have processor idle states with extremely low power consumption](https://msdn.microsoft.com/en-us/ie/dn613899%28v=vs.100%29?f=255&MSPPError=-2147217396)" (<https://msdn.microsoft.com/en-us/ie/dn613899%28v=vs.100%29?f=255&MSPPError=-2147217396>), while [some websites claim that enabling core parking positley influence the Gaming performance](https://windows10gadgets.pro/tipstricks/coreparking.html) (<https://windows10gadgets.pro/tipstricks/coreparking.html>) (higher FPS) and also showing more "benchmark points" in synteical benchmarks.

While some users argue over the benefits of core parking, [Microsoft have chosen to leave the choice of core parking up to the vendors](https://community.amd.com/community/gaming/blog/2017/04/06/amd-ryzen-community-update-3) (<https://community.amd.com/community/gaming/blog/2017/04/06/amd-ryzen-community-update-3>), as they can create there own power plans and shipping with products. With things like Turbo Boost Index and Frequency Scaling Index, a processor is usually able to be controlled more responsively with core parking disabled.

Should I tweak the current Core Parking situation under Windows 10?

First of all, what is Core parking? Basically spoken it allows:

- The processor(s) to go into a "[sleep mode](https://metebalci.com/blog/a-minimum-complete-tutorial-of-cpu-power-management-c-states-and-p-states/)" (<https://metebalci.com/blog/a-minimum-complete-tutorial-of-cpu-power-management-c-states-and-p-states/>).
- Allows your processor(s) to save or to not save energy.

Core parking was first introduced in processor power management (PPM) engine. Your cores are parked that do not have any threads scheduled, and they go into a very low power states when they are not processing interrupts, DPCs, or other strictly affinitized work. The remaining cores are responsible for the remainder of the workload.

The main proplem: On modern multicore processors, the distribution of activity to cores is very dynamic, and overall [helps Turbo Boost](https://stackoverflow.com/a/28921779/103167) (https://stackoverflow.com/a/28921779/103167). Core parking itself acts in very short intervals and does not act long enough for you to perceive any particular core staying parked.

Tweaking can result in more heat?

Yes, in *theory*, if you force specific cores or all cores to stay parked for a very long time it could uneven heating problems insider your processor(s). However, this is depending on how effcent or ineffcent the current power plan is and if the CPU gets enough "room to breath" or not and in which [processor state](https://metebalci.com/blog/a-minimum-complete-tutorial-of-cpu-power-management-c-states-and-p-states/) (https://metebalci.com/blog/a-minimum-complete-tutorial-of-cpu-power-management-c-states-and-p-states/) the CPU currently runs. More heat means means the CPU then starts to throttle and you overall lose performance.

Recommadation:

Let the Windows 10 scheduler spread the activity automatically, this is more effcent and does not create much side-effects (*unless your profile is ineffcent or Windows own scheduler is flawed*).

h3 Hidden Registry command

The registry command was "removed" (it's hidden) and replaced with `powercfg` , as explained over [here](https://superuser.com/a/1437854/900431) (https://superuser.com/a/1437854/900431).

Registry Hive	Registry entry name	Value
:	-----	:-----: :-----:
HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Control\Power\PowerSettings\54533251-82be-4824-96c1-47b60b740d00\0cc5b647-c1df-4637-891a-dec35c318583	Attribute	0 (default is 1)

The official way is now via CMD/PowerShell provided:

- `powercfg -attributes SUB_PROCESSOR CPMINCORES -ATTRIB_HIDE`

This is easier, does not need a reboot and a user is not forced to work with the registry.

h4 Benchmark examples

- [High performance power plan \(Processor rating\)](https://www.userbenchmark.com/UserRun/17287264) (https://www.userbenchmark.com/UserRun/17287264)
- [High performance power plan, with 50% minimum core parking turned on](https://www.userbenchmark.com/UserRun/17287387) (https://www.userbenchmark.com/UserRun/17287387)
- [Balanced power plan \(Processor rating\)](https://www.userbenchmark.com/UserRun/17287490) (https://www.userbenchmark.com/UserRun/17287490)
- [Balanced power plan, with 50% minimum core parking turned on](https://www.userbenchmark.com/UserRun/17287532) (https://www.userbenchmark.com/UserRun/17287532)

Core parking can potentially **increase energy efficiency during lower usage**, which is why it is generally set up on some Windows power plan schemes. Cores are being parked or not depending on configuration of power scheme settings! This will overall affect your energy/performance PC level based on the decision you made (or what was set in PowerCfg).

h4 Power Plans (profiles) & Power Scheme

Power Plans: Windows 10 uses integrated and vendor based power plans to set the Processor Performance which will result in cores being parked or not depending on configuration of power scheme settings.

Power users, vendors (or their applications) can create or "install" their own power plans, which is optional. You can use Microsoft's Provisioning framework to configure the processor power settings described [here](https://docs.microsoft.com/en-us/windows-hardware/customize/power-settings/configure-processor-power-management-options) (https://docs.microsoft.com/en-us/windows-hardware/customize/power-settings/configure-processor-power-management-options). The processor namespace is divided into three sets of identical power processor configurations called power profiles. The power profiles are used by the power processor engine to adapt the performance and parking algorithm on various system use cases.

Power Schemes: They are a collection of settings that controls the power usage of your computer and their internals. Those schemes can be used to e.g. reduce the power consumption of individual devices (e.g. USB/Wifi) or the entire system, see [here](https://docs.microsoft.com/en-us/windows-hardware/customize/power-settings/static-configuration-options-for-core-parking) (https://docs.microsoft.com/en-us/windows-hardware/customize/power-settings/static-configuration-options-for-core-parking) for more information.

Preinstalled power profiles:

- Default** profile is the configuration set that is active most of the time.
- LowLatency** is the profile that is activated during boot and during app launch time.
- LowPower** is the profile that is activated during the buffering phase of media playback scenarios.
- Constrained (*also known as Balanced*)** is a profile activated by the battery saver feature on Windows 10 for desktop editions (Home, Pro, Enterprise, and Education).

PowerCfg profiles several [integrated settings](https://msdn.microsoft.com/en-us/windows/mt422910(v=vs.80)), they are documented over [here](https://docs.microsoft.com/en-us/windows-hardware/customize/power-settings/configure-processor-power-management-options).

- CPMINCORES, CPMINCORES1
- CPMAXCORES, CPMAXCORES1
- LATENCYHINTUNPARK, LATENCYHINTUNPARK1
- PROCTHROTTLEMAX, PROCTHROTTLEMAX1
- PROCTHROTTLEMIN, PROCTHROTTLEMIN1
- PERFINCTHRESHOLD, PERFINCTHRESHOLD1
- PERFINCTIME, PERFINCTIME1
- PERFDECTHRESHOLD, PERFDECTHRESHOLD1
- PERFDECTIME, PERFDECTIME1
- LATENCYHINTPERF, LATENCYHINTPERF1
- PERFAUTONOMOUS
- PERFEPP
- Legacy: [PERFBOOSTMODE](https://docs.microsoft.com/en-us/windows-hardware/customize/power-settings/legacy-config-options-perfboostmode) (ode)
- Legacy: [PERFBOOSTPOL](https://docs.microsoft.com/en-us/windows-hardware/customize/power-settings/legacy-config-options-perfboostpol)

Working with PowerCfr & Windows 10 Power GUI

The command line syntax for PowerCfg is [described over here](https://docs.microsoft.com/en-us/windows-hardware/design/device-experiences/powercfg-command-line-options). You can also create and tweak your own power profile, as shown over [here](https://www.tenforums.com/tutorials/43655-create-custom-power-plan-windows-10-a-post1521767.html#post1521767). The command `powercfg -help` or `powercfg.exe -aliases` typically reveals how to work with the command line syntax and gives you [some](https://www.tenforums.com/tutorials/107613-add-remove-ultimate-performance-power-plan-windows-10-a-2.html) [examples](https://superuser.com/a/1437854/900431).

Reference

- [Core Parking in Windows \(Disable for More Performance\)](https://www.overclock.net/forum/297-general-processor-discussions/1544554-core-parking-windows-disable-more-performance.html)
- [Core Parking Information on Quick CPU Page](https://coderbag.com/product/quickecpu)
- [CPU Parking Options in Power Settings](https://www.tenforums.com/performance-maintenance/36871-cpu-parking-options-power-settings.html)
- [Enable - Disable CPU Turbo Boost from Command Line](https://www.sakya.it/wordpress/enable-disable-cpu-turbo-boost-from-command-line/)
- [How to Add or Remove Maximum processor state from Power Options](https://www.tenforums.com/tutorials/107967-add-remove-maximum-processor-state-power-options-windows.html)
- [Interesting Answer on Microsoft Forum - Core Parking Disabled](https://social.technet.microsoft.com/Forums/en-US/8d6085d8-2c26-426c-87ac-2ba189b77aa5/core-parking-not-working-after-upgrade?forum=win10itprohardware)
- [Metebalci - A Minimum Complete Tutorial of CPU Power Management, C-states and P-states](https://metebalci.com/blog/a-minimum-complete-tutorial-of-cpu-power-management-c-states-and-p-states/)
- [Microsoft - Power and performance tuning](https://docs.microsoft.com/en-us/windows-server/administration/performance-tuning/hardware/power/power-performance-tuning)
- [Microsoft - Processor performance core parking maximum and minimum cores](https://docs.microsoft.com/en-us/windows-server/administration/performance-tuning/hardware/power/power-performance-tuning)
- [Never, Ever Use a Software That Changing the Registry to Un-Park the Core](https://bitsum.com/forum/index.php?topic=3721.0)
- [New power options in Windows 10 for longer battery life on laptops](http://www.rawinfopages.com/tips/2017/08/new-power-options-in-windows-10/)
- [Notebook Review - Improving Battery Life - Enabling Deeper C States](http://forum.notebookreview.com/threads/guide-improving-battery-life-on-windows-enabling-deeper-c-states.815602/)
- [Overclock - Core Parking in Windows \(Disable for more performance\) 2017](https://www.overclock.net/forum/297-general-processor-discussions/1544554-core-parking-windows-disable-more-performance-3.html)
- [Powercfg.exe and Parked CPU Core's](https://forums.battlefield.com/en-us/discussion/96014/powercfg-exe-and-parked-cpu-cores)
- [Reddit - Core Parking and AMD 2017](https://forums.guru3d.com/threads/windows-power-plan-settings-explorer-utility.416058/page-5)
- [Reddit - Unparking Cores \(Good or Bad?\)](https://www.reddit.com/r/windows/comments/8hw0y7/unparking_cores_good_or_bad/)
- [Reddit - Windows 10 Stutter](https://www.reddit.com/r/Windows10/comments/7a0763/standby_memory_issue_causing_stutters_on_creators/)
- [Some CPU Cores Are Parked While Other Active 2008 R2](https://support.microsoft.com/en-au/help/2534356/some-cpu-cores-are-parked-while-other-active-cpu-cores-have-a-heavy-workload)
- [Stack Exchange Network - Why does Windows 10 have CPU core parking disabled?](https://superuser.com/questions/1435110/why-do-es-windows-10-have-cpu-core-parking-disabled)
- [Steam Community - Windows 10 Doesn't Park CPU Cores?](https://steamcommunity.com/discussions/forum/11/1744478429680383517/)

- [Windows 10 Gadgets - Core Parking](https://windows10gadgets.pro/tipstricks/coreparking.html) (https://windows10gadgets.pro/tipstricks/coreparking.html)
- [How does the Windows 10 scheduler deal with Hyper Threading since Core Parking is disabled by default for Intel CPUs?](http://superuser.com/questions/1188967/how-does-the-windows-10-scheduler-deal-with-hyper-threading-since-core-parking-is-disabled-by-default-for-intel-cpus) (http://superuser.com/questions/1188967/how-does-the-windows-10-scheduler-deal-with-hyper-threading-since-core-parking-is-disabled-by-default-for-intel-cpus)

h3 Microsoft

- [Command Line Power Configuration for Microsoft Windows](https://answers.microsoft.com/en-us/windows/forum/windows_10-power/power-plans-missing/9885bdc8-b11f-4722-9e11-423ef77494a9?auth=1&page=5) (https://answers.microsoft.com/en-us/windows/forum/windows_10-power/power-plans-missing/9885bdc8-b11f-4722-9e11-423ef77494a9?auth=1&page=5)
- [Microsoft - Static Core Parking Options](https://docs.microsoft.com/en-us/windows-hardware/customize/power-settings/static-configuration-options-for-core-parking) (https://docs.microsoft.com/en-us/windows-hardware/customize/power-settings/static-configuration-options-for-core-parking)
- [Microsoft Press - Understanding the Windows I/O System](https://www.microsoftpressstore.com/articles/article.aspx?p=2201309&seqNum=7) (https://www.microsoftpressstore.com/articles/article.aspx?p=2201309&seqNum=7)
- [Power and Performance Tuning Windows Server 2016](https://docs.microsoft.com/en-us/windows-server/administration/performance-tuning/hardware/power/power-performance-tuning) (https://docs.microsoft.com/en-us/windows-server/administration/performance-tuning/hardware/power/power-performance-tuning)
- [Static Configuration Options for Heterogeneous Power Scheduling](https://docs.microsoft.com/en-us/windows-hardware/customize/power-settings/static-configuration-options-for-heterogeneous-power-scheduling) (https://docs.microsoft.com/en-us/windows-hardware/customize/power-settings/static-configuration-options-for-heterogeneous-power-scheduling)

h3 Intel

- [Document Number: 335190-003 7th Generation Intel® Processor Families for H Platforms](https://www.intel.com/content/dam/www/public/us/en/documents/datasheets/7th-gen-core-family-mobile-h-processor-lines-datasheet-vol-1.pdf) (https://www.intel.com/content/dam/www/public/us/en/documents/datasheets/7th-gen-core-family-mobile-h-processor-lines-datasheet-vol-1.pdf)
- [Document Number: 335898-001 Intel® Core X-Series Processor Families](https://images-eu.ssl-images-amazon.com/images/I/A1L1jU3HvIL.pdf) (https://images-eu.ssl-images-amazon.com/images/I/A1L1jU3HvIL.pdf)

h3 AMD

- [AMD - Ryzen product in 2017](https://community.amd.com/community/gaming/blog/2017/04/06/amd-ryzen-community-update-3) (https://community.amd.com/community/gaming/blog/2017/04/06/amd-ryzen-community-update-3)
- [TechPowerUp - 1usmus Custom Power Plan for Ryzen 3000 Zen 2 Processors](https://www.techpowerup.com/review/1usmus-custom-power-plan-for-ryzen-3000-zen-2-processors/2.html) (https://www.techpowerup.com/review/1usmus-custom-power-plan-for-ryzen-3000-zen-2-processors/2.html)

h3 Wikipedia

- [Wikipedia - Heterogeneous System Architecture](https://en.wikipedia.org/wiki/Heterogeneous_System_Architecture) (https://en.wikipedia.org/wiki/Heterogeneous_System_Architecture)
- [Wikipedia - Powercfg](https://en.wikipedia.org/wiki/Powercfg) (https://en.wikipedia.org/wiki/Powercfg)

h3 Programs

- [Bitsum - ParkControl](https://bitsum.com/parkcontrol/) (https://bitsum.com/parkcontrol/)
- [Disable CPU Core Parking Utility](https://www.coderbag.com/Programming-C/Disable-CPU-Core-Parking-Utility) (https://www.coderbag.com/Programming-C/Disable-CPU-Core-Parking-Utility)
- [Intel Turbo Boost](https://www.thewindowsclub.com/intel-turbo-boost-technology-max) (https://www.thewindowsclub.com/intel-turbo-boost-technology-max)
- [Quick CPU](https://coderbag.com/product/quickcpu) (https://coderbag.com/product/quickcpu)

h2 How do I measure the overall OS performance?

It is a difficult question because there are a lot of scenarios, hardware combinations & considerations to make. A [benchmark is maybe not reliable](https://www.gamernexus.net/guides/3240-bench-theory-reliability-standard-deviation-of-game-benchmarks) (https://www.gamernexus.net/guides/3240-bench-theory-reliability-standard-deviation-of-game-benchmarks) and you must consider that every benchmark [contains margin of errors](https://www.eurogamer.net/articles/digitalfoundry-2018-the-trouble-with-pc-benchmark-modes) (https://www.eurogamer.net/articles/digitalfoundry-2018-the-trouble-with-pc-benchmark-modes) (which is normal). A synthetic benchmark basically spoken gives you an idea of how well a hardware can perform a given task but the best way to rely upon results is real world usage. Too many things can affect your gaming performance, from part variances within the same product, different test benches e.g. using different CPUs/Motherboards at different settings, different test environment (e.g. temperature), different drivers & settings and sometimes different software versions (patches of games or Windows versions e.g. Build 1809 vs 1903 (No Retpoline vs [Retpoline](https://en.wikipedia.org/wiki/Spectre_(security_vulnerability)#General_approaches) (https://en.wikipedia.org/wiki/Spectre_(security_vulnerability)#General_approaches))). That said do not use in-game [benchmarks](https://en.wikipedia.org/wiki/Benchmark_(computing)) (https://en.wikipedia.org/wiki/Benchmark_(computing)).

I usually use and recommend the [phoronix-test-suite](https://www.phoronix-test-suite.com/) (https://www.phoronix-test-suite.com/) (it's an "real program benchmark") because it does not only benchmark [certain aspects](https://en.wikipedia.org/wiki/Benchmark_(computing)#Challenges) (https://en.wikipedia.org/wiki/Benchmark_(computing)#Challenges) (Kernel, synthetic, component bench), it is also able to run across multiple systems and checks security based parameters to see how the OS "scores". The test itself also includes parameters to show the margin of errors, so it's not an ordinary benchmark tool. It is well documented, open source and gets constantly updated. Keep in mind that you should run it multiple times to get an average overview, however those tools do not help if you have in-game problems because several games and their game engines might have several other issues (which no benchmark can show you). I highly recommend that you run the suite, and then compare how well your favourite game runs in-game, then you have something *objective* to test against.

[CapFrameX](https://github.com/DevTechProfile/CapFrameX) (https://github.com/DevTechProfile/CapFrameX) is a fantastic tool to identify frame drops or G-Sync/FreeSync related issues. The program is for beginners and professionals and can help to check what might cause lags or to get an overview of your hardware and how well it really performs in-game, since it supports DirectX and Vulkan API's it basically can analyze almost every application/game.

h2 System timer resolution

Changing the [OS wide timer](https://docs.microsoft.com/en-us/windows/desktop/api/timeapi/nf-timeapi-timebeginperiod#remarks) (https://docs.microsoft.com/en-us/windows/desktop/api/timeapi/nf-timeapi-timebeginperiod#remarks) can produce [up to 30% performance improvement](https://www.youtube.com/watch?v=_hcuYiqib9I&feature=youtu.be&t=365) (https://www.youtube.com/watch?v=_hcuYiqib9I&feature=youtu.be&t=365) (depends on the game).

Since Windows 10 1603+, the timer resolution is automatically changed every X seconds. This requires you to keep the [timer application](https://vwww.org/contribution/windows-system-timer-tool) (https://vwww.org/contribution/windows-system-timer-tool) open. However, and this is a big one, as soon as a specific application is open that requests another timer, the timer script/tool becomes ineffective. For example, Discord (uses the electron framework - which is known to be [CPU-hungry](https://github.com/electron/electron/issues/11908) (https://github.com/electron/electron/issues/11908)) - if Discord and the timer tool are open at the same time, you will notice that Discord automatically "overwrites" the timings set by the running timing tool.

h3 Application timings

- Run `powercfg /energy` in an elevated command line prompt, this will automatically (after some time [minutes]) create a [power efficiency diagnostics report](https://support.microsoft.com/en-us/help/976034/guided-help-get-a-detailed-power-efficiency-diagnostics-report-for-you) (https://support.microsoft.com/en-us/help/976034/guided-help-get-a-detailed-power-efficiency-diagnostics-report-for-you). Which will be stored under `C:\WINDOWS\system32\energy-report.html`. Open the report via a Browser and search (`CTRL+F`) for `Platform Timer Resolution:Outstanding Timer Request`, this reveals which (running) application/process changes or overwrites the OS timer. It also reveals what resolution the process really requests.

h2 (Disable) Fullscreen optimizations

Fullscreen optimization (the option) was first introduced in Windows 10 Build 1607 ("Anniversary Update"). It attempts to convert games running in the "classic" exclusive fullscreen mode into using borderless window mode with flip model presentation instead.

h3 Benefits

- The optimizations combines the benefits of "classic exclusive fullscreen mode" with borderless window mode, by utilizing Microsoft's flip model presentation in a maximized borderless window.
- Available for DirectX 9+ based games and applications.

h3 Cons & problems

- FSO is known to cause frame pacing issues with some games e.g. Far Cry.

h3 Disabling FSO

While the ability to *system-wide* disable fullscreen optimizations for all the executables was removed in Windows 10 v1803 (April Update), you can do so on a *per-game executable basis* (valid on 1809 and 1903 too): navigate to the game, `Right-click on the executable`, select `Properties` and click on the `Compatibility` tab, check `Disable fullscreen optimizations` box then hit `OK`.

There is an unofficial script avbl. to [automatically "Disable Fullscreen Optimizations"](https://pastebin.com/1twn8geB) (https://pastebin.com/1twn8geB) based on the user choice too. This works on Windows 10 Build 1803 up to Windows Build 1903 too. Some people reporting that it will fix in-game stuttering..

h2 Desktop Window Manager (dwm.exe)

[DWM](https://en.wikipedia.org/wiki/Desktop_Window_Manager) (https://en.wikipedia.org/wiki/Desktop_Window_Manager) (also known as Desktop Compositing Engine or DCE) has in general an *huge* impact on the OS/application performance, it usually *consumes a lot* CPU "power". It exists since Windows Vista and is a part of the Windows OS and since Windows Vista it uses [WDDM](https://en.wikipedia.org/wiki/Windows_Display_Driver_Model) (https://en.wikipedia.org/wiki/Windows_Display_Driver_Model) to share the surface with DWM. In earlier versions it was possible to "disable" DWM, however this is these days not so easy because the process gets restarted every time WinLogon.exe was started.

Disabling DWM is not recommended! But if you want to, you still could do that, [even under Windows 10](https://github.com/Biswa96/Juankyrd/blob/master/c/KillDwm.c) (https://github.com/Biswa96/Juankyrd/blob/master/c/KillDwm.c). The boost is definitely there, however it breaks a lot of other WDDM functions.

h3 Windows Defender and Desktop Window Manager (dwm.exe)

[Another way](https://docs.microsoft.com/en-us/windows/security/threat-protection/windows-defender-exploit-guard/customize-exploit-protection) (https://docs.microsoft.com/en-us/windows/security/threat-protection/windows-defender-exploit-guard/customize-exploit-protection) (but **insecure**) is to disable threat mitigation for the Desktop Window Manager (dwm.exe) itself within Windows Defender. Open Windows Defender (`Settings` -> `Exploit Protection` -> `Program Settings`) and then add `C:\Windows\System32\dwm.exe` to the path, `Check` override system settings for CFG, DEP, EAF, ASLR, SEHOP (vulnerability protections) and ensure it's turned `off`. **It's not recommended to do that!**

h2 Responsiveness and the OS impact

The following things should be as low as possible:

- Audio latency
- Compression
- Ping/Network latency (use the ping command to quickly check your average latency which might be different [depending on several factors])

- DPC Latency (use [LatencyMon](https://www.resplendence.com/latencymon) (https://www.resplendence.com/latencymon) to check it)
- *Overall network latency* (this can't be tested easily because it is depening on several factors)
- Keyboard input processing (depending on yourself + how fast the keyboard/mouse + OS processes the signals)
- Security holes - [MDS, ZombieLoad, Spectre/Meltdown, L1TF etc.](https://www.phoronix.com/scan.php?page=article&item=mds-zombieload-mit&num=1) (https://www.phoronix.com/scan.php?page=article&item=mds-zombieload-mit&num=1).

The following things should be as high as possible:

- The energy profile (should be set to Ultimate), see [here](https://docs.microsoft.com/en-us/windows-hardware/customize/desktop/customize-power-slider) (https://docs.microsoft.com/en-us/windows-hardware/customize/desktop/customize-power-slider) why.
- Frequency (only in case you do Overclocking (!)) of several components CPU/GPU (requires testing to stabilize it).
- Power Management Mode (nVidia Control Panel [NVCP]) "application controlled" is enough.

h2 Choose the "right" Windows 10 edition

Why does it matter? [Different Windows versions](https://en.wikipedia.org/wiki/Windows_10_editions) (https://en.wikipedia.org/wiki/Windows_10_editions) come with different preinstalled apps or programs. Practical you could remove them but this costs a) your time and b) you might not be able to remove all apps (*easily*). The Backgrounds apps and started services matter in a sense that they might result in a [higher CPU/RAM usage](http://www.blackviper.com/service-configurations/) (http://www.blackviper.com/service-configurations/) or lags because these apps and services typically trying to update themselves in the *right moment* which might even ends-up with a [Bluescreen \(BSOD\)](https://answers.microsoft.com/en-us/windows/forum/all/windows-10-update-causing-blue-screen-errors-daily/45a08401-87c9-4c52-b160-f8548bc42c6f) (https://answers.microsoft.com/en-us/windows/forum/all/windows-10-update-causing-blue-screen-errors-daily/45a08401-87c9-4c52-b160-f8548bc42c6f) or [crashes](https://answers.microsoft.com/en-us/windows/forum/apps/windows_10-outlook_mail/windows-update-caused-loss-of-mail-app-windows/aec58c9c-9371-46b4-ab69-2b54353eff5c) (https://answers.microsoft.com/en-us/windows/forum/apps/windows_10-outlook_mail/windows-update-caused-loss-of-mail-app-windows/aec58c9c-9371-46b4-ab69-2b54353eff5c).

- Avoid Preview or any Insider builds
- Avoid Pro or Home Editions
- Use Education (Enterprise) or LTSC versions (*if possible*)

Edu./Ent. or LTSC versions have no pre-installed apps OR/AND you can take control over them without the need to rely on third-party apps or script. Group policy Editor (GPO - gpedit.msc) and the integrated Windows 10 own Settings can control every app/store/update behavior.

h2 Driver installation

The driver installation order *does matter* and might help to solve or prevent some problems.

- OS
- Chipset (reboot) - You do not need to "update" regularly your chipset driver, just install the missing driver (inf) component. There is no benefit in "updating" the existent driver.
- USB (e.g. ASMedia Technology Inc.)
- Audio/GPU (the order doesn't matter)
- Monitor/keyboard/mouse etc.

Why is the specific order needed? Some driver(s) and their installers having troubles detecting your current hardware if e.g. the chipset isn't installed first, other programs simply conflicting with the IRQ order (older mobos).

It's not necessary to re-install the entire OS in case you made a mistake, I suggest you use [DDU](https://www.wagnardsoft.com/) (https://www.wagnardsoft.com/) in combination with [RAPR](https://github.com/lostindark/DriverStoreExplorer) (https://github.com/lostindark/DriverStoreExplorer) in order to get rid of *possible* leftovers and old/redundant drivers in Windows driver store folder. at this point I also suggest to create a "Driver" folder on another HDD/SSD/USB drive and place all drivers you got in there. It's maybe the first time a bit more effort to collect all of them but it helps to update the drivers + keeps on eye on the current installed ones and possible (in case you store the last two versions) you have a revision in case the newer drivers are problematic.

h2 Reasons to avoid WUS Drivers

In [some situations](https://www.windowscentral.com/amd-marks-50-years-gold-edition-radeon-vii-ryzen-7-bundles) (https://www.windowscentral.com/amd-marks-50-years-gold-edition-radeon-vii-ryzen-7-bundles) the offered drivers from Windows Update Servers (WUS) are outdated compared to the ones which the OEM (AMD, nVidia, Intel etc.) offers which means that they can cause some performance or even crash related issue. My own *advice* is to use [Wumgr](https://github.com/DavidXanatos/wumgr) (https://github.com/DavidXanatos/wumgr) which is an open source program to take control over Windows Updates (in case you don't want to mess with GPO's/Registry). Windows 10 Build 1903+ got an function to manually install drivers via the integrated GUI (in this case you might not need Wumgr).

Even some OEMs might have outdated drivers (listed on the driver pages e.g. Realtek), Station-Drivers or Win-Raid Forum often have more up-2-dated drivers.

I recommend:

- Create a folder on a usb drive or external drive where you store all you drivers (in case you re-install Windows and to keep track of your current installed drivers).

- Label the folders and files correctly which shows you directly which version you currently have installed.

h4 Tools like Snappy Driver Installer & Co

Snappy Driver ([Origin](https://www.snappy-driver-installer.org/) (<https://www.snappy-driver-installer.org/>) - the version which is open-source and without ads) can help you in case your OEM doesn't provide a manual to identify your current hardware, however in my experience such tools should be avoided because they *maybe* install or identify the wrong hardware which might only ends-up with more problems and they could contain ads or are infected with malware (since they are a huge target because driver based malware is harder to remove/spot and the installer might execute the installation process under administrative rights to gain access). My *advice* is (even if it's more effort) to [manually identify the installed hardware](https://www.windowscentral.com/how-check-your-computer-full-specifications-windows-10) (<https://www.windowscentral.com/how-check-your-computer-full-specifications-windows-10>) create ther driver folder and download/verify the driver from trusted websites sich as Station-Drivers & Win-RAID.

h2 Install AiO runtimes or repacks such as DirectX, NetFramework or Visual C++ Redistributable

There is a wide-spreading myth that says that it helps to install repacked versions to *gain gaming performance*. This is wrong and there was never any evidence given, but it's true that such repacks a) are smaller in download size (compared to the original packages) b) helping you to avoid wasting SSD/HDD space on your OS partition. For example you can use [abbodi1406 vcredist package](https://github.com/abbodi1406/vcredist) (<https://github.com/abbodi1406/vcredist>) which only installs the latest Microsoft Visual C++ Redistributable Runtimes which comes without any payload - in other words you save a lot of space. While this in theory is a good thing it is not always practical because some Game Stores such as Steam or even some applications which you install forcing you to *re-install* those packages because they won't probably detect that you already have those runtimes installed. This basically means you override such a repack with the official MS packages and at the end you won't gain anything out of it.

Problems:

- Repacks could contain malware
- Steam and other stores, or even some applications will override or force you to install the official runtimes even if newer ones are already installed
- No performance benefits

h2 Operating System (general tweaking advice)

- Switch unneeded services (services.msc) from automatic to manual start. This lowers CPU/Ram usage and migrates possible attack scenarios e.g. it prevents NetBios attacks (if that service was disabled). Keep in mind that some services can't be disabled (depending on the Windows 10 Build) which then require you to disable it via registry.
- Disable unneeded tasks in Task Scheduler, pretty much as same as above the goal is to reduce CPU/Ram consumption and to possible migrate attack scenarios.
- Disable unneeded logs in Event Viewer - This is more or less optional, performance wise this doesn't gain anything but it helps your SSD life since Windows writes those events in the background to a file/disk.
- Do not install any AntiVirus product, if you like to use an AV then stick with [Windows Defender because it's good enough](http://www.tomsguide.com/us/windows-defender,review-2209.html) (<http://www.tomsguide.com/us/windows-defender,review-2209.html>) as proven in 'independent' AV tests.
- Use only a good NAT Firewall and [Sandboxie](https://www.sandboxie.com/) (<https://www.sandboxie.com/>) (paid product), sandbox untrusted or unknown stuff. Sandboxie has no performance drops unless you use it 'cracked' or load huge apps into it. Windows 10 Build 1903/20H1 will get it's own *free* but limited sandbox function which would be a free alternative to Sandboxie.
- Change [Core Parking only in XP, 7 and 8](https://social.technet.microsoft.com/Forums/azure/en-US/76dac4e8-ce8f-4b83-b33d-bbef50ae5d9c/cpu-core-parking-in-windows-7-should-it-be-left-alone-or-should-users-disable-it?forum=w7itproperf) (<https://social.technet.microsoft.com/Forums/azure/en-US/76dac4e8-ce8f-4b83-b33d-bbef50ae5d9c/cpu-core-parking-in-windows-7-should-it-be-left-alone-or-should-users-disable-it?forum=w7itproperf>) because since Windows 10 the OS itself controls it automatically already.
- Delete unneeded UWP apps and uninstall the stuff you won't need, this gains some SSD/HDD space, reduces memory/cpu usage (since such apps mostly running in the background or trying to update themselves).

h3 Windows 7 vs Windows 10

Gaming on Windows 10 is not necessary better or worst, it depends on several factors such as the API used by the game developer, the engine itself, whether the game was "designed" and "optimized" for Windows 10/7, the drivers and driver profiles, and more.



(<https://www.youtube.com/watch?v=RkHFYKDOo74>)

h3 Windows 10 LTSC (Build 1607) Vs. LTSC (2019)



(<https://youtu.be/ghNe876N3TY>)

h2 Application performance impact

Some applications and their integrated drivers from external devices like [a mouse can cause serious FPS drops](https://old.reddit.com/r/Doom/comments/8a1m9s/psa_deactivate_the_new_razer_chroma_option_in/) (https://old.reddit.com/r/Doom/comments/8a1m9s/psa_deactivate_the_new_razer_chroma_option_in/). Keep in mind that you should update the driver ASAP once there is a new one out, this might not only fix performance problems, it often also fixes security holes, simply review the changelog manually before you apply the changes to see if it's worth it and do backups before you install a new upgrade.

h3 Fix Memory Issue with Windows 10 Build 1607 (Anniversary Update)

- Download [EmptyStandbyList](https://wj32.org/wp/software/empty-standby-list/) (<https://wj32.org/wp/software/empty-standby-list/>) and put it under e.g. C:\ (ensure you don't move this file)
- Right click > Properties and select 'Run as Admin' under compatibility.
- Open your Task Scheduler > Create Task (on the far right).
- General Tab (give it a name, doesn't matter which one!). Under security options > Change user or group > Advanced > Find Now > go down and choose SYSTEM (important to make it run silently in the background). Tick 'Run with highest privileges' and 'Hidden' at the bottom. You can find an example [here](https://stackoverflow.com/questions/6568736/how-do-i-set-a-windows-scheduled-task-to-run-in-the-background) (<https://stackoverflow.com/questions/6568736/how-do-i-set-a-windows-scheduled-task-to-run-in-the-background>).
- Click Triggers Tab > New > On a schedule > One Time. Tick repeat task every 5 minutes (possibly excessive but it causes no issues). Also choose 'for the duration of: indefinitely'
- Actions tab > Start A program > Point to the EmptyStandbyList.exe file.
- The Standby memory is automatically cleared every 5 mins.

h2 Task Scheduler & Anti-Telemetry script

List of Windows tasks to disable (privacy & performance related):

- \Microsoft\Windows\Application Experience > AitAgent, ProgramDataUpdater
- \Microsoft\Windows\Autochk > Proxy
- \Microsoft\Windows\Customer Experience Improvement Program > Consolidator, KernelCeipTask, UsbCeip
- \Microsoft\Windows\DiskDiagnostic > Microsoft-Windows-DiskDiagnosticDataCollector
- \Microsoft\Windows\Maintenance > WinSAT
- \Microsoft\cSystemRestore > SR
- \Microsoft\Windows\WindowsBackup > ConfigNotification

- \Microsoft\Windows Defender > MP Scheduled Scan
- \Library\Microsoft\Windows\WindowsColorSystem\Calibration Loader (disable it if you use your own Display Color Profile)
- \Microsoft\Microsoft\CDPUserSvc (see [here](https://account.microsoft.com/privacy/activity-history) (https://account.microsoft.com/privacy/activity-history))

Copy the following text and save it as `antitelemetry.cmd` .

```
>-

@echo off
%windir%\system32\reg.exe query "HKU\S-1-5-19" 1>nul 2>nul || goto :eof
echo.
echo Anti-Telemetry blocking script
echo Some of the commands may return error(s) - you can ignore them!
reg add HKLM\SOFTWARE\Policies\Microsoft\Windows\WindowsUpdate /v DisableOSUpgrade /t REG_DWORD /d 1 /f >nul 2>&1
reg add "HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\WindowsUpdate\Auto Update" /f /v IncludeRecommendedUpdates /t REG_DWORD /d 0 >nul 2>&1
reg delete "HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\WindowsUpdate\OSUpgrade" /f >nul 2>&1
reg add HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\WindowsUpdate\OSUpgrade /v AllowOSUpgrade /t REG_DWORD /d 0 /f >nul 2>&1
reg add "HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\AppCompatFlags\Appraiser" /v HaveUploadedForTarget /t REG_DWORD /d 1 /f >nul 2>&1
reg add "HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\AppCompatFlags\AIT" /v AITenable /t REG_DWORD /d 0 /f >nul 2>&1
reg delete "HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\AppCompatFlags\ClientTelemetry" /f >nul 2>&1
reg add "HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\AppCompatFlags\ClientTelemetry" /v DontRetryOnError /t REG_DWORD /d 1 /f >nul 2>&1
reg add "HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\AppCompatFlags\ClientTelemetry" /v IsCensusDisabled /t REG_DWORD /d 1 /f >nul 2>&1
reg add "HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\AppCompatFlags\ClientTelemetry" /v TaskEnableRun /t REG_DWORD /d 1 /f >nul 2>&1
reg delete "HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\AppCompatFlags" /v UpgradeEligible /f >nul 2>&1
reg delete "HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\AppCompatFlags\Appraiser" /f >nul 2>&1
reg delete "HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\AppCompatFlags\TelemetryController" /f >nul 2>&1
reg add HKLM\SOFTWARE\Microsoft\SQMClient\IE /v CEIPenable /t REG_DWORD /d 0 /f >nul 2>&1
reg add HKLM\SOFTWARE\Microsoft\SQMClient\IE /v SqmLoggerRunning /t REG_DWORD /d 0 /f >nul 2>&1
reg add HKLM\SOFTWARE\Microsoft\SQMClient\Reliability /v CEIPenable /t REG_DWORD /d 0 /f >nul 2>&1
reg add HKLM\SOFTWARE\Microsoft\SQMClient\Reliability /v SqmLoggerRunning /t REG_DWORD /d 0 /f >nul 2>&1
reg add HKLM\SOFTWARE\Microsoft\SQMClient\Windows /v DisableOptinExperience /t REG_DWORD /d 1 /f >nul 2>&1
reg add HKLM\SOFTWARE\Microsoft\SQMClient\Windows /v CEIPenable /t REG_DWORD /d 0 /f >nul 2>&1
reg add HKLM\SOFTWARE\Microsoft\SQMClient\Windows /v SqmLoggerRunning /t REG_DWORD /d 0 /f >nul 2>&1
sc.exe config DiagTrack start= disabled >nul 2>&1
sc.exe stop DiagTrack >nul 2>&1
reg delete HKLM\SYSTEM\ControlSet001\Control\WMI\AutoLogger\AutoLogger-Diagtrack-Listener /f >nul 2>&1
reg delete HKLM\SYSTEM\ControlSet001\Control\WMI\AutoLogger\Diagtrack-Listener /f >nul 2>&1
reg delete HKLM\SYSTEM\ControlSet001\Control\WMI\AutoLogger\SQMLogger /f >nul 2>&1
reg delete HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Diagnostics\DiagTrack /f >nul 2>&1
reg delete HKLM\SOFTWARE\Policies\Microsoft\Windows\DataCollection /f >nul 2>&1
reg add HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Diagnostics\DiagTrack /v DiagTrackAuthorization /t REG_DWORD /d 0 /f >nul 2>&1
takeown /f %ProgramData%\Microsoft\Diagnosis /A /r /d y >nul 2>&1
icacls %ProgramData%\Microsoft\Diagnosis /grant:r *S-1-5-32-544:F /T /C >nul 2>&1
del /f /q %ProgramData%\Microsoft\Diagnosis\*.rbs >nul 2>&1
del /f /q /s %ProgramData%\Microsoft\Diagnosis\ETLLogs\* >nul 2>&1
schtasks /Change /TN "\Microsoft\Windows\Application Experience\Microsoft Compatibility Appraiser" /DISABLE >nul 2>&1
schtasks /Change /TN "\Microsoft\Windows\Application Experience\ProgramDataUpdater" /DISABLE >nul 2>&1
schtasks /Change /TN "\Microsoft\Windows\Application Experience\AitAgent" /DISABLE >nul 2>&1
schtasks /Change /TN "\Microsoft\Windows\PerfTrack\BackgroundConfigSurveyor" /DISABLE >nul 2>&1
schtasks /Delete /F /TN "\Microsoft\Windows\Application Experience\Microsoft Compatibility Appraiser" >nul 2>&1
schtasks /Delete /F /TN "\Microsoft\Windows\Application Experience\ProgramDataUpdater" >nul 2>&1
schtasks /Delete /F /TN "\Microsoft\Windows\Application Experience\AitAgent" >nul 2>&1
schtasks /Delete /F /TN "\Microsoft\Windows\PerfTrack\BackgroundConfigSurveyor" >nul 2>&1
TIMEOUT /T 5
```

To ensure the blocking settings are always set eg. after an upgrade/KB update, you can create a schedule task to run the script at logon as example:

1. Copy the above mentioned script into `C:\Windows` and then open command prompt as administrator and execute:
2. `SCHTASKS /Create /F /RU "SYSTEM" /RL HIGHEST /SC ONSTART /TN BlockW10 /TR "cmd /c %windir%\antitelemetry.cmd"`

h2 Random stuttering

h5 The following workaround is specifically for Windows 1703 >= 1803

Some games might have random stutter because Windows tries to free some resources (more or less good) which might causes game stutters. A more detailed explanation can be found to the issue over [here](https://forums.guru3d.com/threads/fix-game-stutter-on-win-10-1703-1803.420251/page-12#post-5590635) (https://forums.guru3d.com/threads/fix-game-stutter-on-win-10-1703-1803.420251/page-12#post-5590635). Possible workarounds in form from batch files or small utilities can be found [here](https://forums.guru3d.com/threads/fix-game-stutter-on-win-10-1703-1803.420251/) (https://forums.guru3d.com/threads/fix-game-stutter-on-win-10-1703-1803.420251/).

h3 Power Management

Set your power setting to maximum performance (it's not needed to use an "ultimate" power-plan) if you graphic card driver has an 'maximum performance' settings enable it too, this will ensure that the GPU uses it's full potential.

h3 GameDVR

Whenever you stream games via OBS Studio, make sure you disable GameDVR, it is also a smart advice to in general stay away from it since most applications & games are simply not optimized for it - this doesn't have anything to do with that MS *fucked something up* it's more that some older games (and even several new ones) are not adopted to get any benefit in such *low-latency* mode.

Keep in mind that the registry tweak is not needed in case you're on RS5+ because **Settings > Gaming > Game bar** is the way to go (since some people reporting to have other registry values).

Windows Registry Editor Version 5.00

[HKEY_CURRENT_USER\Software\Microsoft\GameBar]

```
"UseNexusForGameBarEnabled"=dword:00000000
```

"AutoGameModeEnabled"=dword:00000000

[HKEY_CURRENT_USER\System\GameConfigStore]

"GameDVR_Enabled"=dword:00000000

"GameDVR_FSEBehavior"=dword:00000002

"GameDVR_FSEBehaviorMode"=dword:00000002

"GameDVR_HonorUserFSEBehaviorMode"=dword:00000000

"GameDVR_DXGIHonorFSEWindowsCompatible"=dword:00000000

"GameDVR_EFSEFeatureFlags"=dword:00000000

```
"Win32_AutoGameModeDefaultProfile"=hex:02,00,01,00,00,00,c4,20,00,00,00,00,00,\
```

00,\

00,00	
-------	--

```
"Win32_GameModeRelatedProcesses"=hex:01,00,01,00,01,00,c0,00,c6,02,50,54,c7,02,\
```

70,00,61,00,6e,00,65,00,6c,00,2e,00,65,00,78,00,65,00,00,00,8c,00,4e,8d,e1,\

74,b8,ed,d2,02,18,4c,c7,02,1e,00,00,00,b8,ed,d2,02,1e,00,00,00,0f,00,00,00,\

30,e7,00,00,00,00,00,00,00,00,00,00,00,00,00,00,00,00,00,00,00,\

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Since Windows 1803+ you don't need to disable GameDVR anymore, **only in case you have game specific issues!**, in such case use the provided workaround, or better report it to the game developer studio to address it directly.

GameBarPresenceWriter.exe

GameBarPresenceWriter.exe is a part of the Xbox Game Bar function and typically runs in the background. The process can't be killed because it will automatically be restarted. You can use [this script](https://github.com/CHEF-KOCH/GamingTweaks/tree/master/Tools/GameBarPresenceWriter) (https://github.com/CHEF-KOCH/GamingTweaks/tree/master/Tools/GameBarPresenceWriter) to kill it once and for all.

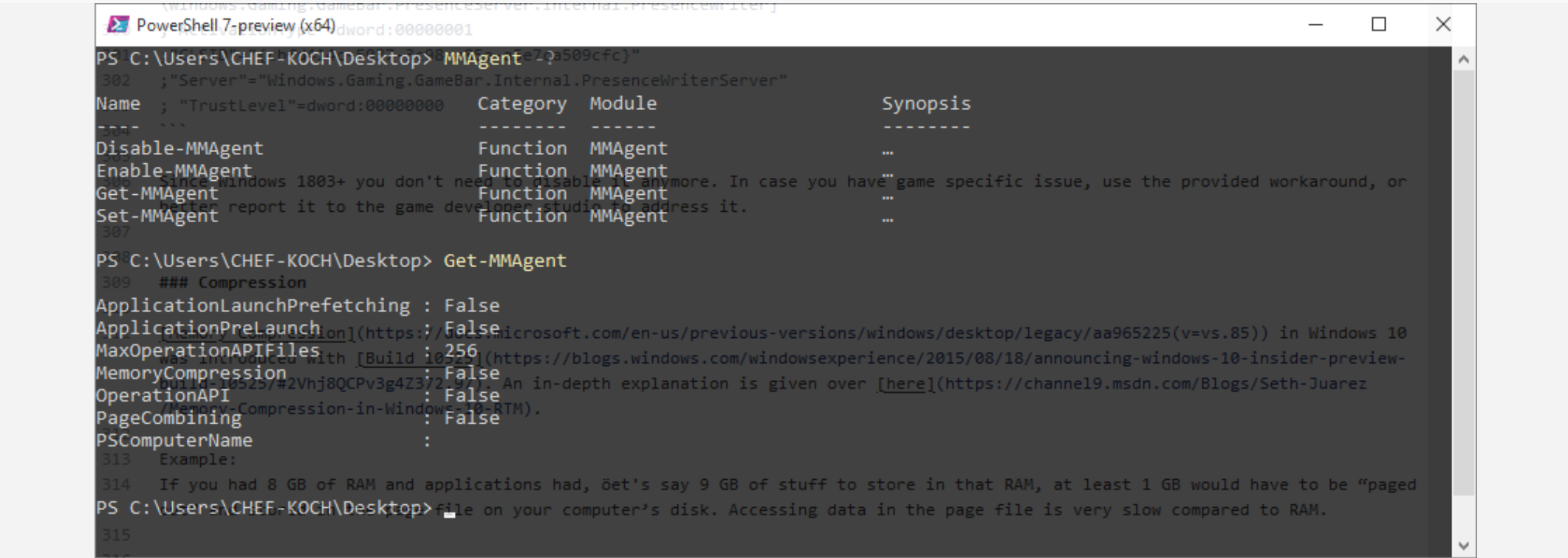
h3 Windows 10 Memory Compression

[Memory Compression](https://docs.microsoft.com/en-us/previous-versions/windows/desktop/legacy/aa965225(v=vs.85)) (https://docs.microsoft.com/en-us/previous-versions/windows/desktop/legacy/aa965225(v=vs.85)) in Windows 10 was introduced with [Build 10525](https://blogs.windows.com/windowsexperience/2015/08/18/announcing-windows-10-insider-preview-build-10525/#2Vhj8QCPv3g4Z372.97) (https://blogs.windows.com/windowsexperience/2015/08/18/announcing-windows-10-insider-preview-build-10525/#2Vhj8QCPv3g4Z372.97). An in-depth explanation is given over [here](https://channel9.msdn.com/Blogs/Seth-Juarez/Memory-Compression-in-Windows-10-RTM) (https://channel9.msdn.com/Blogs/Seth-Juarez/Memory-Compression-in-Windows-10-RTM).

Example:
If you had 8 GB of RAM and applications had, let's say 9 GB of stuff to store in that RAM, at least 1 GB would have to be “paged out” and stored in the page file on your computer’s disk. Accessing data in the page file is very slow compared to RAM. Overall said, it all comes down to the amount of RAM. In theory you should not notice if compression is enabled or disabled on a system with 16+ GB, the sad truth is that some applications and games are not well "optimized" and disabling the compression algo might result in an overall *smoother* application/game behavior, especially if you run many UWP apps in the background.

Open a PowerShell with Administrator privileges then, run the command `Disable-MMAgent -MemoryCompression` and reboot. This disables the compression. To enable it later again, run again powershell as admin and run the `Enable-MMAgent -MemoryCompression` command. Some task managers can show memory compression information like Windows own task-manager, [ProcessHacker](https://wj32.org/processhacker/forums/viewtopic.php?t=2421) (https://wj32.org/processhacker/forums/viewtopic.php?t=2421) and [Process Explorer](https://www.howtogeek.com/319933/what-is-memory-compression-in-windows-10/) (https://www.howtogeek.com/319933/what-is-memory-compression-in-windows-10/).

For more details, see [this post](https://github.com/CHEF-KOCH/GamingTweaks/issues/26) (https://github.com/CHEF-KOCH/GamingTweaks/issues/26).



h2 OS specific functions and their impact on the overall game performance

h3 Windows 10 Game Mode

[Game Mode](https://support.xbox.com/en-US/games/game-setup/use-game-mode-gaming-on-pc) (https://support.xbox.com/en-US/games/game-setup/use-game-mode-gaming-on-pc) - Game Mode allows the system to prioritize CPU and GPU resources to the game running in the foreground, and aims to provide an increase of overall frame rate, or peaks, as well as an increase of average frame rate, or consistency. In the latest versions of Windows starting with Build 1809 & 1903, Game Mode no longer does GPU/CPU prioritization! This means (background) processes get more resources in case other applications e.g. OBS Studio starving your other (free) resources to death (e.g. because encoding).

- Enabling Game Mode stops/prevents Windows Update from updating drivers and stops it from notifying you of non-critical updates.
- It forces (if enabled) to use minimum of 100% CPU resources on Desktop systems (not laptops!) in order to help reduce CPU fluctuations which may cause performance issues. It does exactly the same thing as doing it manually by going to Processor power management and setting `Minimum processor state` manually to 100% (high performance power plan).
- Game mode **is not** enabled by default, this is based on the used Windows version. Starting with 1809+ the game itself can decide to enable (use) Game Mode or not. The supported game list will automatically been updated, based if xyz game does utilize Game Mode.
- Game Mode works different on laptops to avoid overheating issues.

That been said, turn on Game Mode in 1903+ builds **does not hurt** the gaming performance anymore. The statement: "It prioritizes the game process and suspends windows system processes as much as possible." is not valid anymore on newer Windows builds.

Fullscreen Optimizations

[Disable fullscreen optimizations for an application](https://support.microsoft.com/en-us/help/15078/windows-make-older-programs-compatible) (https://support.microsoft.com/en-us/help/15078/windows-make-older-programs-compatible) - The Fullscreen Optimizations for Apps and Games feature is enabled by default in Windows 10 which is suppose to enhance your gaming experience by prioritizing your CPU and GPU resources to your games and apps.

Retpoline

[Retpoline](https://www.zdnet.com/article/microsoft-rolls-out-googles-retpoline-spectre-mitigation-to-windows-10-users/) (https://www.zdnet.com/article/microsoft-rolls-out-googles-retpoline-spectre-mitigation-to-windows-10-users/) - Basically spoken a new mitigation method against [Spectre variant 2](https://en.wikipedia.org/wiki/Spectre_(security_vulnerability)) (https://en.wikipedia.org/wiki/Spectre_(security_vulnerability)) by Google, however [Retpoline is not applicable to Skylake and later processors from Intel](https://news.ycombinator.com/item?id=16072775) (https://news.ycombinator.com/item?id=16072775). It can have an [huge impact on game performance](https://www.techspot.com/news/79038-microsoft-introduces-drastic-performance-fix-spectre-variant.html) (https://www.techspot.com/news/79038-microsoft-introduces-drastic-performance-fix-spectre-variant.html). It is still *not entirely* fixed.

Disable Windows Update

- Disable driver updates delivered through Windows Update - Microsoft can deliver an outdated or even in worse case scenarios deliver a wrong driver which then has an impact on your overall OS performance. I suggest to manually install all drivers and keep an eye on them.

Memory Compression

[Memory Compression](https://www.techrepublic.com/article/how-to-monitor-memory-compression-in-windows-10/) (https://www.techrepublic.com/article/how-to-monitor-memory-compression-in-windows-10/) - Disabling memory compression can [increase application stability while using demanding games for CPU](https://www.dobreprogramy.pl/deton24/Dlaczego-moje-gry-sie-zacinaja-diagnostyka-Ustawianie-pliku-wymiany-ReadyBoost-Testy-trybu-gry-i-kompresji-pamieci-w-Windows--porady,85501.html) (https://www.dobreprogramy.pl/deton24/Dlaczego-moje-gry-sie-zacinaja-diagnostyka-Ustawianie-pliku-wymiany-ReadyBoost-Testy-trybu-gry-i-kompresji-pamieci-w-Windows--porady,85501.html). Some games (not all) benefit from the memory compression enabled.

Control Flow Guard (CF-Guard)

[Disable "Control Flow Guard"](https://docs.wholetomato.com/default.asp?W790) (https://docs.wholetomato.com/default.asp?W790) - In general you should never disable this optional on a global basis, however [some applications](https://windowsreport.com/windows-10-windows-10-cfg-slow-browser/) (https://windowsreport.com/windows-10-windows-10-cfg-slow-browser/) can get a slight boost (e.g. faster loading times) when disabled. My advice is to leave it on.

Force CPU affinity

[Force CPU affinity](https://support.microsoft.com/en-ph/help/296930/registry-values-for-process-control-parameters) (https://support.microsoft.com/en-ph/help/296930/registry-values-for-process-control-parameters) - You don't need programs like ProcessLasso for this, you can use [ProcessHacker](https://processhacker.sourceforge.io/) (https://processhacker.sourceforge.io/) or Microsoft's own [ProcessExplorer](https://docs.microsoft.com/en-us/sysinternals/downloads/process-explorer) (https://docs.microsoft.com/en-us/sysinternals/downloads/process-explorer). The Windows own [Task Manager does not store the tweak permanently](https://i.imgur.com/wCOwRZS.png) (https://i.imgur.com/wCOwRZS.png) (!). It needs to be reapplied each time the process launches. Keep in mind that this tweak is questionable it is depending on the application.

General OC & stability

- Turn off all power features in your current BIOS settings. **No EIST, C1, C3, C6, etc.** Lock the CPU (cores) at one speed, and keep it there otherwise you might see BSOD's or other application related crashes during your tests/benchmarks!
- (*optional* - to test the stability) Rerun Windows System Assessment Tool (WinSAT) after updating your drivers, changing hardware or overclocking via 'winsat formal -restart clean'.
- Setting **dynamictick** & **tscsyncpolicy** might helps to increase the stability. An official explanation is given over [here](https://docs.microsoft.com/en-us/windows-hardware/drivers/devtest/bcdedit--set) (https://docs.microsoft.com/en-us/windows-hardware/drivers/devtest/bcdedit--set).

```
>_

//Enable dynamictick & tscsyncpolicy

bcdedit /set useplatformclock true

bcdedit /set tscsyncpolicy Enhanced

bcdedit /set disabledynamictick yes

// Remove dynamictick & tscsyncpolicy (OS default)

bcdedit /deletevalue useplatformclock

bcdedit /deletevalue tscsyncpolicy

bcdedit /deletevalue disabledynamictick
```

h2 AMD specific

h4 RAM OC on Zen

Overclocking isn't just about getting more MHz it also means and requires that you understand how your hardware works. RAM overclocking is critical because it might instantly results in crashes, BSOD's or freezes when done wrong. Reading the correct values is mandatory and helps to find the *best OC potential* and also helps with the CPU limitations, especially on AM4- & TR4-systems.

- Download [DRAM Calculator for Ryzen utility](https://www.techpowerup.com/download/ryzen-dram-calculator/) (https://www.techpowerup.com/download/ryzen-dram-calculator/) - it's free
- Follow the [step-by-step guide](https://www.techpowerup.com/reviews/AMD/Ryzen_Memory_Tweaking_Overclocking_Guide/) (https://www.techpowerup.com/reviews/AMD/Ryzen_Memory_Tweaking_Overclocking_Guide/)
- (*optional*) Taiphoon Burner tool (see reference) is a useful free program to read-out your current DRAM configuration, which can be exported as HTML-Report. You can import the file into DRAM Calculator in order to start with the correct values. This helps to calculate a more specific OC-Profile based on your RAM-Kit.

h4 Boost (Turbo)

[AMD confirmed an firmware issue](https://twitter.com/AMDRyzen/status/1168901636162539536?ref_src=twsrc%5Etfw) (https://twitter.com/AMDRyzen/status/1168901636162539536?ref_src=twsrc%5Etfw) with the (latest) AMD Ryzen-3000-CPU's (Family: "Matisse"). This problem lead into a reduced turbo clock speed. A firmware update (Bios update v46.44.00). [AGESA Combo-AM4 1.0.0.4](https://www.reddit.com/r/Amd/comments/dtgutp/an_update_on_the_am4_platform_agesa_1004/) (https://www.reddit.com/r/Amd/comments/dtgutp/an_update_on_the_am4_platform_agesa_1004/) fixes all known issue.

AM4 Platform Updates

AMD has recently released a new AGESA to manufacturers, version 1004. With over 150 changes, this is a significant milestone release in the development of the AM4 platform. We wanted to share some background in support of our release and particularly in advance of the AMD Ryzen 9 3950X processor launch on Nov. 25th.

AGESA 1004 Schedule

AGESA 1004 is primarily focused on stability, and here are the essentials you need to know:

This is the only AGESA validated and recommended for the AMD Ryzen 9 3950X. Your motherboard must have a BIOS with AGESA 1004 for full and proper support.

It adds final support for "Eco Mode," accessible via Ryzen Master. Eco Mode brings AMD Configurable TDP (cTDP) technology to 3rd Gen Ryzen. With one button, you can run a 95W/105W model at 65W, or 65W models at 45W. This feature has been especially fun to use with the 3900X – 12 cores at 65W! We know some of you have been very interested in a configuration like this, and now you can have it with the touch of a button.

AGESA 1004 is the "big one" that reunites the codebase for all AM4-compatible processors. Up until now, 3rd Gen Ryzen has been a fork of the mainline. So, if you have an older Ryzen CPU and have been waiting for a major AGESA release: this is it! This will allow motherboard vendors to unify their CPU and chipset support lists on one codebase.

As a major stability release, you're undoubtedly curious about the major changes. Here are some of the big ones:

- AMD X570 stability and compatibility with add-in devices.
- Improvements to interoperability of PCIe, USB, SATA, and device reset capabilities.
- Additional enhancements for PCIe device support and stability.
- Improved system stability when switching through ACPI power states.
- Improved POST/boot times. This will vary by motherboard.

There are also two performance enhancements:

- improvements to fastest core utilization, and further boost frequency
- optimizations for the AMD Ryzen 9 3900X.

Because of the size and complexity of AGESA 1004, and the extensive regression testing required on older motherboards (unified codebase), please expect the QA and release schedule to take a few weeks yet for older motherboards. For newer motherboards, like AMD X570, releases have already started to hit the web over the past 2 weeks.

In summary, lots of goodness coming in AGESA 1004 and we look forward to hearing your feedback. Thanks for reading — We really appreciate it.

Reference & Tools

- [AMD Ryzen Memory Tweaking & Overclocking Guide](https://www.techpowerup.com/reviews/AMD/Ryzen_Memory_Tweaking_Overclocking_Guide/) (https://www.techpowerup.com/reviews/AMD/Ryzen_Memory_Tweaking_Overclocking_Guide/)
- [HCI MemTest](https://hcdesign.com/memtest/) (https://hcdesign.com/memtest/)
- [Taiphooon Burner](http://www.softnology.biz/) (http://www.softnology.biz/)

RAM & Windows game stutter

On AMD Threadripper systems you can enable Dynamic Local Mode, which acts like something which prioritize the RAM (needs a restart). In the future [AMD is going to do this with a software automatically in the background](https://community.amd.com/community/gaming/blog/2018/10/05/previewing-dynamic-local-mode-for-the-amd-ryzen-threadripper-wx-series-processors) (https://community.amd.com/community/gaming/blog/2018/10/05/previewing-dynamic-local-mode-for-the-amd-ryzen-threadripper-wx-series-processors). Dynamic Local Mode (DLM) can be controlled via a Windows Service (on/off) and the needed update in order to get this automatically controlled will be introduced October 29. 2018 with an chipset driver update (and is for now an [Windows exclusive function](https://en.m.wikipedia.org/wiki/SCHED_DEADLINE) (https://en.m.wikipedia.org/wiki/SCHED_DEADLINE) because Linux already handles it better).

General Game stuttering (every xyz seconds)

- Check your OS Event Reporter tools for signs of hardware failures
- Check the PSU (and the rest of your hardware components)
- Is your Sound card/GPU/SSD/M.2 in the correct slot? 16x and not 1x/4x?
- Have you checked your PSU again?

Assuming that your hardware is 100% perfect and working you can use one of these tools in order to workaround the Windows integrated compression bug, see [here](https://github.com/CHEF-KOCH/GamingTweaks/tree/master/Tools/RAM) (https://github.com/CHEF-KOCH/GamingTweaks/tree/master/Tools/RAM).

Meltdown/MDS/ZombieLoad Performance Impact (MUST READ!)

- [Looking At The Linux Performance Two Years After Spectre / Meltdown Mitigations \(phoronix.com\)](https://www.phoronix.com/scann.php?page=article&item=spectre-meltdown-2&num=1) (https://www.phoronix.com/scann.php?page=article&item=spectre-meltdown-2&num=1)
- [The Performance Impact Of MDS / Zombieload Plus The Overall Cost Now Of Spectre/Meltdown/L1TF/MDS \(phoronix.com\)](https://www.phoronix.com/scan.php?page=article&item=mds-zombieload-mit&num=1) (https://www.phoronix.com/scan.php?page=article&item=mds-zombieload-mit&num=1)
- [RAMBleed \(rambleed.com\)](https://rambleed.com/) (https://rambleed.com/)
- [Network Cache ATtack \(short: NetCAT\) \(vusec.net\)](https://www.vusec.net/projects/netcat/) (https://www.vusec.net/projects/netcat/) - [CVE-2019-11184](http://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2019-11184) (http://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2019-11184)

NVIDIA's FPS limiters (V1, V2 & V3) vs RTSS vs In-engine

- [Comprehensive benchmarking of NVIDIA's FPS limiters \(V1, V2 & V3\) vs RTSS vs In-engine: A research of their frame time consistency and approximate input lag.Software \(self.allbenchmarks\)](https://old.reddit.com/r/allbenchmarks/comments/fbuk9x/comprehensive_benchmarking_of_nvidias_fps/) (https://old.reddit.com/r/allbenchmarks/comments/fbuk9x/comprehensive_benchmarking_of_nvidias_fps/)

Hardware explained

- [DDR4 vs GDDR5/GDDR6 vs LPDDR4 vs HBM2: Different Memory \(RAM\) Types Explained \(hardwaretimes.com\)](https://www.hardwaretimes.com/ddr4-vs-gddr5-gddr6-vs-lpddr4-vs-hbm2-different-memory-ram-types-explained/) (https://www.hardwaretimes.com/ddr4-vs-gddr5-gddr6-vs-lpddr4-vs-hbm2-different-memory-ram-types-explained/)

Windows Defender Exploit Guard Configuration

- [Configure Windows Defender ExploitGuard, Reset all ProcessMitigation, Import clean recommended Baseline Configuration \(github.com\)](https://github.com/gunnarhaslinger/Windows-Defender-Exploit-Guard-Configuration) (https://github.com/gunnarhaslinger/Windows-Defender-Exploit-Guard-Configuration)

HPET

- [AMD and Intel Have Different HPET Guidance \(anandtech.com\)](https://www.anandtech.com/show/12678/a-timely-discovery-examining-amd-2nd-gen-ryzen-results/3) (https://www.anandtech.com/show/12678/a-timely-discovery-examining-amd-2nd-gen-ryzen-results/3)
- [AMD recommended HPET be disabled \(realworldtech.com\)](https://www.realworldtech.com/forum/?threadid=176149&curpostid=176160) (https://www.realworldtech.com/forum/?threadid=176149&curpostid=176160)
- [BCDEdit /set \(docs.microsoft.com\)](https://docs.microsoft.com/en-us/windows-hardware/drivers/devtest/bcdedit--set?redirectedfrom=MSDN) (https://docs.microsoft.com/en-us/windows-hardware/drivers/devtest/bcdedit--set?redirectedfrom=MSDN)
- [Common Stutter / Lag / Crash Fix \(HPET Guide\) \(linustechtips.com\)](https://linustechtips.com/main/topic/915727-common-stutter-lag-crash-fix-hpet-guide/) (https://linustechtips.com/main/topic/915727-common-stutter-lag-crash-fix-hpet-guide/)
- [Disable High Precision Event Timer \(HPET\) in BIOS \(forums.guru3d.com\)](http://forums.guru3d.com/showpost.php?p=3770113&postcount=2) (http://forums.guru3d.com/showpost.php?p=3770113&postcount=2)

- [Disable HPET\(High Precision Event Timer\) \(mikemartin.co\)](https://www.mikemartin.co/system_guides/hardware/motherboard/disable_high_precision_event_timer_hpet) (https://www.mikemartin.co/system_guides/hardware/motherboard/disable_high_precision_event_timer_hpet)
- [Disable or enable High Precision Event Timer\(HTEP\)? \(bleepingcomputer.com\)](https://www.bleepingcomputer.com/forums/t/677082/disable-or-enable-high-precision-event-timerhtep/) (https://www.bleepingcomputer.com/forums/t/677082/disable-or-enable-high-precision-event-timerhtep/)
- [Dynamic Tick Mode explained \(doc.micrium.com\)](https://doc.micrium.com/display/osiidoc/Dynamic+Tick+Mode) (https://doc.micrium.com/display/osiidoc/Dynamic+Tick+Mode)
- [High Precision Event Timer \(wikipedia.org\)](https://en.wikipedia.org/wiki/High_Precision_Event_Timer) (https://en.wikipedia.org/wiki/High_Precision_Event_Timer)
- [How To Improve Gaming Performance By Disabling HPET Settings? \(silicophilic.com\)](https://silicophilic.com/disabling-hpet-setting_s/) (https://silicophilic.com/disabling-hpet-setting_s/)
- [The Clock Tick \(doc.micrium.com\)](https://doc.micrium.com/display/osiidoc/The+Clock+Tick) (https://doc.micrium.com/display/osiidoc/The+Clock+Tick)
- [Windows Timestamp \(windowstimestamp\)](https://www.windowstimestamp.com/description) (https://www.windowstimestamp.com/description)

h2 General Research

- [A MINIMUM COMPLETE TUTORIAL OF CPU POWER MANAGEMENT, C-STATES AND P-STATES \(metebalci.com\)](https://metebalci.com/blog/a-minimum-complete-tutorial-of-cpu-power-management-c-states-and-p-states/) (https://metebalci.com/blog/a-minimum-complete-tutorial-of-cpu-power-management-c-states-and-p-states/)
- [ASUS PEG Link Mode Guide Rev. 2.0 \(techarp.com\)](http://www.techarp.com/showarticle.aspx?artno=248&pgno=0) (http://www.techarp.com/showarticle.aspx?artno=248&pgno=0)
- [Better on the inside: under the hood of Windows 8 \(arstechnica.com\)](https://arstechnica.com/information-technology/2012/10/better-on-the-inside-under-the-hood-of-windows-8/2/) (https://arstechnica.com/information-technology/2012/10/better-on-the-inside-under-the-hood-of-windows-8/2/)
- [Catalyst Antialiasing Guide \(beyond3d.com\)](http://www.beyond3d.com/content/articles/37/1) (http://www.beyond3d.com/content/articles/37/1)
- [Does Core i7 Hyper-Threading Helps? \(vr-zone.com\)](http://vr-zone.com/articles/does-core-i7-hyper-threading-helps-/6160.html?doc=6160) (http://vr-zone.com/articles/does-core-i7-hyper-threading-helps-/6160.html?doc=6160)
- [Nvidia Inspector profile\(s\) to fix Low Power for some applications \(github.com\)](https://github.com/Ociidii-Works/Nvidia-GPU-Low-Power-Fix) (https://github.com/Ociidii-Works/Nvidia-GPU-Low-Power-Fix)
- [The Gigabyte GA-X48T-DQ6 - Redefining the High End? \(anandtech.com\)](http://www.anandtech.com/show/2410/13) (http://www.anandtech.com/show/2410/13)

h2 Windows specific

- [Demystifying Fullscreen Optimizations \(devblogs.microsoft.com\)](https://devblogs.microsoft.com/directx/demystifying-full-screen-optimizations/) (https://devblogs.microsoft.com/directx/demystifying-full-screen-optimizations/)
- [Microsoft educates Insiders on how Windows 10 handles memory \(onmsft.com\)](https://www.onmsft.com/news/microsoft-educates-insiders-windows-10-handles-memory) (https://www.onmsft.com/news/microsoft-educates-insiders-windows-10-handles-memory)
- [Windows 10 1809 Performance Impact \(ict-r.com\)](https://www.ict-r.com/windows-10-1809-performance-impact/) (https://www.ict-r.com/windows-10-1809-performance-impact/)
- [Windows Vista's SuperFetch and ReadyBoost Analysed \(tomshardware.com\)](https://www.tomshardware.com/reviews/windows-vista-superfetch-and-readyboostanalyzed,1532-2.html) (https://www.tomshardware.com/reviews/windows-vista-superfetch-and-readyboostanalyzed,1532-2.html)

h2 Mouse & Keyboard

- [Mouse Sensitivity \(mouse-sensitivity.com\)](https://www.mouse-sensitivity.com/) (https://www.mouse-sensitivity.com/)
- [Wireless Mouse Click Latency Analysis \(gamersnexus.net\)](https://www.gamersnexus.net/guides/2594-wireless-mouse-click-latency-analysis-vs-wired) (https://www.gamersnexus.net/guides/2594-wireless-mouse-click-latency-analysis-vs-wired)

h2 Dithering under Windows

- [Disable temporal dithering on Windows - Amulet Hotkey \(resources.amulethotkey.com\)](https://resources.amulethotkey.com/download/KBA_144_Disable_temporal_dithering_on_Windows_May_2018.pdf) (https://resources.amulethotkey.com/download/KBA_144_Disable_temporal_dithering_on_Windows_May_2018.pdf)

h2 NetCode, Networking

- [SYN cookies FAQ \(cr.yp.to\)](http://cr.yp.to/syncookies.html) (http://cr.yp.to/syncookies.html)

h2 Frametimes, FPS

- [GeDoSaTo FPS capping, modding controversies \(blog.metaclassofnil.com\)](http://blog.metaclassofnil.com/?p=715) (http://blog.metaclassofnil.com/?p=715)

h2 TFT

- [TFT Central](https://www.tftcentral.co.uk/) (https://www.tftcentral.co.uk/)

h2 SSD

- [TheSSDReview](http://www.thessdreview.com/) (http://www.thessdreview.com/)
- [The real and complete story - Does Windows defragment your SSD? \(hanselman.com\)](https://www.hanselman.com/blog/TheRealAndCompleteStoryDoesWindowsDefragmentYourSSD.aspx) (https://www.hanselman.com/blog/TheRealAndCompleteStoryDoesWindowsDefragmentYourSSD.aspx)

h2 General Game tweak-knowledge

- [PC Gaming Wiki \(pcgamingwiki.com\)](https://pcgamingwiki.com/wiki/Home) (https://pcgamingwiki.com/wiki/Home) - Game configs, suggestions & tricks database (similar like Wikipedia just for games)
- [WSGF Forum \(wsgf.org\)](http://www.wsgf.org/forums/viewforum.php?f=64&sid=07479fb019fa626648795c596de5e9ea) (http://www.wsgf.org/forums/viewforum.php?f=64&sid=07479fb019fa626648795c596de5e9ea) - Game Mods, discussions and requests.

h2 YouTube Channel with focus of game analysis

- [Battle\(non\)sense](https://www.youtube.com/channel/UCP7QY6L5pvmm0-stL-pNFrw) (https://www.youtube.com/channel/UCP7QY6L5pvmm0-stL-pNFrw) (*net code and latency*)

h2 YouTube Channel with focus of hardware analysis

- [JayzTwoCents](https://www.youtube.com/user/Jayztwocents) (https://www.youtube.com/user/Jayztwocents) (*all sorts of hardware*)
- [Gamers Neus](https://www.youtube.com/channel/UChIs72whgZI9w6d6FhwGGHA) (https://www.youtube.com/channel/UChIs72whgZI9w6d6FhwGGHA) (*all sorts of hardware*)
- [BitWid](https://www.youtube.com/channel/UCftcLVz-jtPXoH3cWUUDwYw) (https://www.youtube.com/channel/UCftcLVz-jtPXoH3cWUUDwYw) (*all sorts of hardware*)
- [TechSource](https://www.youtube.com/channel/UChIZGfcnjHI0DG4nweWEduw) (https://www.youtube.com/channel/UChIZGfcnjHI0DG4nweWEduw) (*all sorts of hardware*)
- [Rocket Jump Ninja](https://www.youtube.com/channel/UCGJaDZC7PChgd-XMwcbZkiw) (https://www.youtube.com/channel/UCGJaDZC7PChgd-XMwcbZkiw) (*mice*)

h2 Game Emulators

- [RetroArch \(retroarch.com\)](https://www.retroarch.com/) (https://www.retroarch.com/)
- [Mesen high-accuracy NES and Famicom \(mesen.ca\)](https://www.mesen.ca/) (https://www.mesen.ca/)
- [Snes9x \(github.com\)](https://github.com/snes9xgit/snes9x) (https://github.com/snes9xgit/snes9x)
- [Higan \(byuu.org\)](https://www.byuu.org/emulation/higan/) (https://www.byuu.org/emulation/higan/)
- [Emulator accuracy tests \(tasvideos.org\)](http://tasvideos.org/EmulatorResources.html#AccuracyTests) (http://tasvideos.org/EmulatorResources.html#AccuracyTests)

h2 Raytracing benchmarks

- [Crytek Neon Noir](https://www.cryengine.com/marketplace/product/neon-noir#) (https://www.cryengine.com/marketplace/product/neon-noir#)

h2 Forums

- [Official nVidia Forum](https://forums.geforce.com/) (https://forums.geforce.com/)
- [Official AMD Forum](https://community.amd.com/community/support-forums/general-discussion) (https://community.amd.com/community/support-forums/general-discussion)
- [Official Intel Forum](https://forums.intel.com/s/?language=en_US) (https://forums.intel.com/s/?language=en_US)
- [Guru3d Forum](https://forums.guru3d.com/) (https://forums.guru3d.com/)

h2 DRM Performance Impact

- [A non misleading benchmark of Denuvo in AC Origins](https://www.reddit.com/r/CrackWatch/comments/e6p6kc/a_non_misleading_benchmark_of_denuvo_in_ac_origins/) (https://www.reddit.com/r/CrackWatch/comments/e6p6kc/a_non_misleading_benchmark_of_denuvo_in_ac_origins/) + [A 4c/4t benchmark of Denuvo in AC Origins \[Follow up\]](https://www.reddit.com/r/CrackWatch/comments/e73xb8/a_4c4t_benchmark_of_denuvo_in_ac_origins_follow_up/) (https://www.reddit.com/r/CrackWatch/comments/e73xb8/a_4c4t_benchmark_of_denuvo_in_ac_origins_follow_up/)
- [Resident Evil 2 Remake is faster by 4-12fps without the Denuvo anti-tamper tech \(dsogaming.com\)](https://www.dsogaming.com/news/resident-evil-2-remake-is-faster-by-4-12fps-without-the-denuvo-anti-tamper-tech/) (https://www.dsogaming.com/news/resident-evil-2-remake-is-faster-by-4-12fps-without-the-denuvo-anti-tamper-tech/)
- [Denuvo Really Does Cripple PC Gaming Performance \(https://www.extremetech.com\)](https://www.extremetech.com/gaming/282924-denuvo-really-does-cripple-pc-gaming-performance) (https://www.extremetech.com/gaming/282924-denuvo-really-does-cripple-pc-gaming-performance)
- [Devil May Cry 5 PC's Denuvo DRM has a CPU hit \(eurogamer.net\)](https://www.eurogamer.net/articles/digitalfoundry-2019-devil-may-cry-5-pc-denuvo-protection-tested) (https://www.eurogamer.net/articles/digitalfoundry-2019-devil-may-cry-5-pc-denuvo-protection-tested) - ([Updated test](https://www.techpowerup.com/reviews/Denuvo/Performance_Loss_Test/3.html) (https://www.techpowerup.com/reviews/Denuvo/Performance_Loss_Test/3.html))

h2 Other common DRM problems

- [Tron: Evolution SecuROM DRM expiration makes game unplayable 9 years after release \(community.pcgamingwiki.com\)](https://community.pcgamingwiki.com/topic/4476-tron-evolution-securom-drm-expiration-makes-game-unplayable-9-years-after-release) (https://community.pcgamingwiki.com/topic/4476-tron-evolution-securom-drm-expiration-makes-game-unplayable-9-years-after-release/)

h2 Power Management

- [Device power management reference \(docs.microsoft.com\)](https://docs.microsoft.com/en-us/windows-hardware/drivers/kernel/device-power-management-reference) (https://docs.microsoft.com/en-us/windows-hardware/drivers/kernel/device-power-management-reference)

h2 Kernel Latencies

- [User Mode Versus Kernel Mode \(docs.microsoft.com\)](https://docs.microsoft.com/en-us/windows-hardware/drivers/audio/user-mode-versus-kernel-mode) (https://docs.microsoft.com/en-us/windows-hardware/drivers/audio/user-mode-versus-kernel-mode)
- [Effectively Measure and Reduce Kernel Latencies for Real-time Constraints \(elinux.org\)](https://elinux.org/images/a/a9/ELC2017-_Effectively_Measure_and_Reduce_Kernel_Latencies_for_Real-time_Constraints_%281%29.pdf) (https://elinux.org/images/a/a9/ELC2017-_Effectively_Measure_and_Reduce_Kernel_Latencies_for_Real-time_Constraints_%281%29.pdf)
- [Analysis of High Resolution Timer Latency Using Kernel Analysis System in Embedded System \(ieeexplore.ieee.org\)](https://ieeexplore.ieee.org/document/4804584) (https://ieeexplore.ieee.org/document/4804584)
- [Acquiring high-resolution time stamps \(docs.microsoft.com\)](https://docs.microsoft.com/en-us/windows/desktop/sysinfo/acquiring-high-resolution-time-stamps) (https://docs.microsoft.com/en-us/windows/desktop/sysinfo/acquiring-high-resolution-time-stamps)
- [Example 15: Measuring DPC/ISR Time](https://docs.microsoft.com/en-us/windows-hardware/drivers/devtest/example-15--measuring-dpc-isr-time) (https://docs.microsoft.com/en-us/windows-hardware/drivers/devtest/example-15--measuring-dpc-isr-time)

h2 PhysX

- [Announcing NVIDIA PhysX SDK 5.0 \(news.developer.nvidia.com\)](https://news.developer.nvidia.com/announcing-nvidia-physx-sdk-5-0/) (https://news.developer.nvidia.com/announcing-nvidia-physx-sdk-5-0/)

h2 Audio Latency in Windows

- [Low Latency Audio \(docs.microsoft.com\)](https://docs.microsoft.com/en-us/windows-hardware/drivers/audio/low-latency-audio) (https://docs.microsoft.com/en-us/windows-hardware/drivers/audio/low-latency-audio)

- [WavePci Latency \(docs.microsoft.com\)](https://docs.microsoft.com/en-us/windows-hardware/drivers/audio/wavepci-latency) (https://docs.microsoft.com/en-us/windows-hardware/drivers/audio/wavepci-latency)
- [WaveCyclic Latency \(docs.microsoft.com\)](https://docs.microsoft.com/en-us/windows-hardware/drivers/audio/wavecyclic-latency) (https://docs.microsoft.com/en-us/windows-hardware/drivers/audio/wavecyclic-latency)
- [Stream Latency During Recording \(docs.microsoft.com\)](https://docs.microsoft.com/en-us/windows-hardware/drivers/audio/stream-latency-during-recording) (https://docs.microsoft.com/en-us/windows-hardware/drivers/audio/stream-latency-during-recording)

h2 For Developers

- [Guidelines for Writing DPC Routines \(docs.microsoft.com\)](https://docs.microsoft.com/en-us/windows-hardware/drivers/kernel/guidelines-for-writing-dpc-routines) (https://docs.microsoft.com/en-us/windows-hardware/drivers/kernel/guidelines-for-writing-dpc-routines)
- [Managing Input/Output for Drivers \(docs.microsoft.com\)](https://docs.microsoft.com/en-us/windows-hardware/drivers/kernel/managing-input-output-for-drivers?redirectedfrom=MSDN) (https://docs.microsoft.com/en-us/windows-hardware/drivers/kernel/managing-input-output-for-drivers?redirectedfrom=MSDN)
- [Scheduling Priorities \(docs.microsoft.com\)](https://docs.microsoft.com/en-us/windows/win32/procthread/scheduling-priorities?redirectedfrom=MSDN) (https://docs.microsoft.com/en-us/windows/win32/procthread/scheduling-priorities?redirectedfrom=MSDN)

h2 Driver & Firmware Resources

- [8BitDo Firmware page \(support.8bitdo.com\)](http://support.8bitdo.com/) (http://support.8bitdo.com/)
- [Station-Drivers \(station-drivers.com\)](http://www.station-drivers.com/) (http://www.station-drivers.com/)
- [Standalone Steam Controller open source driver \(github.com\)](https://github.com/ynsta/steamcontroller) (https://github.com/ynsta/steamcontroller)
- [Super Nt Official Firmware \(support.analogue.co\)](https://support.analogue.co/hc/en-us/articles/360000557452-Super-Nt-Firmware-Update-v4-7) (https://support.analogue.co/hc/en-us/articles/360000557452-Super-Nt-Firmware-Update-v4-7)
- [Super Nt Jailbreak \(github.com\)](https://github.com/SmokeMonsterPacks/Super-NT-Jailbreak) (https://github.com/SmokeMonsterPacks/Super-NT-Jailbreak)
- [Nt Mini Firmware Links \(support.analogue.co\)](https://support.analogue.co/hc/en-us/articles/115001947108-Nt-mini-Firmware-Update-v2-3) (https://support.analogue.co/hc/en-us/articles/115001947108-Nt-mini-Firmware-Update-v2-3)
- [Nt Mini Firmware Jailbreak \(blog.kevtris.org\)](http://blog.kevtris.org/blogfiles/ntm_firmware_verJB2.0.zip) (http://blog.kevtris.org/blogfiles/ntm_firmware_verJB2.0.zip)
- [Mega Sg firmware \(support.analogue.co\)](https://support.analogue.co/hc/en-us/articles/360024836892-Mega-Sg-Firmware-Update-v4-2) (https://support.analogue.co/hc/en-us/articles/360024836892-Mega-Sg-Firmware-Update-v4-2)
- [Mega SG Jailbreak \(github.com\)](https://github.com/SmokeMonsterPacks/Mega-Sg-Jailbreak) (https://github.com/SmokeMonsterPacks/Mega-Sg-Jailbreak)

h2 Legendary RAM OC Guides

- [RAM-Overclocking und Auswirkungen auf Spiele bei Intel CPUs \(computerbase.de\)](https://www.computerbase.de/forum/threads/intel-core-i-serie-ram-overclocking-auswirkungen-auf-spiele.1849970/) (https://www.computerbase.de/forum/threads/intel-core-i-serie-ram-overclocking-auswirkungen-auf-spiele.1849970/)
- [Intel Skylake-X \(Socket 2066\) OC-Guide \(hardwareluxx.de\)](https://www.hardwareluxx.de/community/f139/intel-skylake-x-socket-2066-oc-guide-1172969.html) (https://www.hardwareluxx.de/community/f139/intel-skylake-x-socket-2066-oc-guide-1172969.html)
- [AMD Ryzen - RAM OC Community \(computerbase.de\)](https://www.computerbase.de/forum/threads/amd-ryzen-ram-oc-community.1829356/) (https://www.computerbase.de/forum/threads/amd-ryzen-ram-oc-community.1829356/)

h2 Comprehensive AMD Ryzen Memory & OC articles

- [The AnandTech Guide to Video Card Overclocking Software](https://www.anandtech.com/show/8589/anandtech-guide-to-video-card-overclocking-software/2) (https://www.anandtech.com/show/8589/anandtech-guide-to-video-card-overclocking-software/2)
- [AMD Ryzen Memory Tweaking & Overclocking Guide \(techpowerup.com\)](https://www.techpowerup.com/reviews/AMD/Ryzen_Memory_Tweaking_Overclocking_Guide/) (https://www.techpowerup.com/reviews/AMD/Ryzen_Memory_Tweaking_Overclocking_Guide/)
- [AMD RYZEN Overclocking Guide \(overclockers.com\)](https://www.overclockers.com/amd-ryzen-overclocking-guide/) (https://www.overclockers.com/amd-ryzen-overclocking-guide/)
- [Official Corsair Ryzen 3000 Series Memory Overclocking Guide \(corsair.com\) \(.pdf\)](https://www.corsair.com/corsairmedia/sys_master/productcontent/Ryzen3000_MemoryOverclockingGuide.pdf) (https://www.corsair.com/corsairmedia/sys_master/productcontent/Ryzen3000_MemoryOverclockingGuide.pdf)
- [Robert Hallock explains memory and overclocking on AMD Ryzen \(youtube.com\)](https://www.youtube.com/watch?v=vZgpHTaQ10k&feature=youtu.be&t=3m35s) (https://www.youtube.com/watch?v=vZgpHTaQ10k&feature=youtu.be&t=3m35s)

h2 Comprehensive Intel "Coffee Lake" Memory & OC articles

- [The Intel 9th Gen Review: Core i9-9900K, Core i7-9700K and Core i5-9600K Tested \(anandtech.com\)](https://www.anandtech.com/show/13400/intel-9th-gen-core-i9-9900k-i7-9700k-i5-9600k-review) (https://www.anandtech.com/show/13400/intel-9th-gen-core-i9-9900k-i7-9700k-i5-9600k-review)
- [Intel's Core i9-9900K CPU reviewed \(techreport.com\)](https://techreport.com/review/34192/intel-core-i9-9900k-cpu-reviewed) (https://techreport.com/review/34192/intel-core-i9-9900k-cpu-reviewed)

h2 Message Signaled Interrupts

- [Message Signaled Interrupts in PCI/PCIE : Advantages over INTx \(blog.sasken.com\)](https://blog.sasken.com/message-signaled-interrupts-in-pci-pcie-advantages-over-intx) (https://blog.sasken.com/message-signaled-interrupts-in-pci-pcie-advantages-over-intx)

h2 AMD (AGESA) BIOS overview & download links

- [AMD AGESA Bios collections thread \(computerbase.de\)](https://www.computerbase.de/forum/threads/sammelthread-amd-agesa-uebersicht-aller-bios-versionen.1902983/) (https://www.computerbase.de/forum/threads/sammelthread-amd-agesa-uebersicht-aller-bios-versionen.1902983/) - Overview of all AGESA BIOS updates, the website is in german but it's easy to understand, just follow your vendor and mobo and check if there is a link on the right side of the chart or not.

h2 Controller guides

- [Connect the Nintendo Switch Pro Controller to Windows](https://www.gameaccessory.com/post/how-to-connect-your-pro-controller-to-your-pc-infographic) (https://www.gameaccessory.com/post/how-to-connect-your-pro-controller-to-your-pc-infographic)

h2 Programs

h3 Full Screen Optimizations

- [Compatibility Manager \(github.com\)](https://github.com/Skymirrh/CompatibilityManager) (https://github.com/Skymirrh/CompatibilityManager)

h3 USB Polling tools

- [USB polling precision explained \(overclock.net\)](https://www.overclock.net/forum/375-mice/1550666-usb-polling-precision.html) (https://www.overclock.net/forum/375-mice/1550666-usb-polling-precision.html)
- ~~(old) Mouse testser tool (overclock.net)~~ (https://www.overclock.net/forum/375-mice/1535687-mousetester-software.html) & [source code \(github.com\)](https://github.com/microe1/MouseTester) (https://github.com/microe1/MouseTester)
- ~~(new) Mouse tester tool (overclock.net)~~ (https://www.overclock.net/forum/375-mice/1590569-mousetester-software-reloaded.html) & [source code \(github.com\)](http://github.com/dobragab/MouseTester) (http://github.com/dobragab/MouseTester)
- [Mouse movement recorder by MarkC \(donewmouseaccel.blogspot.com\)](https://donewmouseaccel.blogspot.com/) (https://donewmouseaccel.blogspot.com/) & [Source Code](https://github.com/CHEF-KOCH/GamingTweaks/tree/master/Tools/Mouse%20Movement%20Recorder/MouseMovementRecorder%20Source%20Code) (https://github.com/CHEF-KOCH/GamingTweaks/tree/master/Tools/Mouse%20Movement%20Recorder/MouseMovementRecorder%20Source%20Code)
- [Mouse Rate Checker \(softpedia.com\)](http://www.softpedia.com/get/System/System-Miscellaneous/Mouse-Rate-Checker.shtml) (http://www.softpedia.com/get/System/System-Miscellaneous/Mouse-Rate-Checker.shtml)
- [1000hz - Overclock/Underclock your old mouse and increase its polling rate \(github.com\)](https://github.com/vadash/1000hz) (https://github.com/vadash/1000hz)

h3 HPET / QPC Benchmark tools

- [TimerBench](https://www.overclockers.at/downloads/projects/TimerBench%201.4.exe) (https://www.overclockers.at/downloads/projects/TimerBench%201.4.exe) ([source](https://www.overclockers.at/number-crunching/timerbench-ein-benchmark-fuer-windows-timer_249630/page_7) (https://www.overclockers.at/number-crunching/timerbench-ein-benchmark-fuer-windows-timer_249630/page_7))

h3 Frametime capture and analysis tools

- [CapFrameX \(github.com\)](https://github.com/DevTechProfile/CapFrameX) (https://github.com/DevTechProfile/CapFrameX)

h3 Controller Emulators (*for Windows*)

- [DualShock 4 for Windows \(DS4Windows\) \(github.com\)](https://github.com/Ryochan7/DS4Windows) (https://github.com/Ryochan7/DS4Windows)
- [Xbox 360 Controller Emulator \(x360ce.com\)](https://www.x360ce.com/) (https://www.x360ce.com/)

h3 Monitor "OC" programs

- [Custom Resolution Utility \(CRU\) \(monitortests.com\)](http://www.monitortests.com/forum/Thread-Custom-Resolution-Utility-CRU) (http://www.monitortests.com/forum/Thread-Custom-Resolution-Utility-CRU)
- [Pixelclock calculator \(monitortests.com\)](https://www.monitortests.com/pixelclock.php) (https://www.monitortests.com/pixelclock.php)

h3 nVidia Driver "adjustment" programs

- [nVidia Inspector \(download.orbmu2k.de\)](http://download.orbmu2k.de/files/nvidiaInspector.zip) (http://download.orbmu2k.de/files/nvidiaInspector.zip) + [source code](https://github.com/Orbmu2k/nvidiaProfileInspector) (https://github.com/Orbmu2k/nvidiaProfileInspector) (latest final version) + [Latest beta](https://ci.appveyor.com/project/Orbmu2k/nvidiaprofileinspector) (https://ci.appveyor.com/project/Orbmu2k/nvidiaprofileinspector)
- [NV RGBFullRangeToggle \(blog.metaclassofnil.com\)](http://blog.metaclassofnil.com/wp-content/uploads/2012/08/NV_RGBFullRangeToggle.zip) (http://blog.metaclassofnil.com/wp-content/uploads/2012/08/NV_RGBFullRangeToggle.zip) Use this tool if you don't see any 'Output dynamic range' option in the NVCP.

h3 Driver Uninstaller (leftover removal) programs

- [Display Driver Uninstaller \(wagnardsoft.com\)](https://www.wagnardsoft.com/) (https://www.wagnardsoft.com/)

h3 GPU OC programs (AMD/Intel/both)

- [MSI Afterburner + Guide \(forums.guru3d.com\)](https://forums.guru3d.com/threads/overclocking-with-msi-afterburner-at-users-risk.327291/) (https://forums.guru3d.com/threads/overclocking-with-msi-afterburner-at-users-risk.327291/)
- [AMD Memory Tweak \(github.com\)](https://github.com/Eliovp/amdmemorytweak) (https://github.com/Eliovp/amdmemorytweak)
- [Sapphire TriXX OC Utility \(sapphiretech.com\)](https://www.sapphiretech.com/en/software) (https://www.sapphiretech.com/en/software)
- [EVGA Precision XOC/X/X1 \(evga.com\)](https://www.evga.com/precisionxoc/) (https://www.evga.com/precisionxoc/)

h3 AMD CPU "OC" programs

- [AMD Ryzen Master \(download.amd.com\)](https://download.amd.com/Desktop/AMD-Ryzen-Master.exe) (https://download.amd.com/Desktop/AMD-Ryzen-Master.exe) + [Official FAQ](https://www.amd.com/en/support/kb/faq/gpu-775) (https://www.amd.com/en/support/kb/faq/gpu-775)
- [AmdMsrTweaker \(github.com\)](https://github.com/mpollice/AmdMsrTweaker) (https://github.com/mpollice/AmdMsrTweaker)

h3 AMD RAM "OC" programs

- [Ryzen Timing Checker \(techpowerup.com\)](https://www.techpowerup.com/download/ryzen-timing-checker/) (https://www.techpowerup.com/download/ryzen-timing-checker/)
- [DRAM Calculator for Ryzen \(techpowerup.com\)](https://www.techpowerup.com/download/ryzen-dram-calculator/) (https://www.techpowerup.com/download/ryzen-dram-calculator/)

h3 Intel "OC" programs

- ~~Intel® Extreme Tuning Utility - Intel® XTU~~ (https://downloadcenter.intel.com/en/download/24075/Intel-Extreme-Tuning-Utility-Intel-XTU) *Use other programs because the driver might cause performance problems.* Prefer (if possible) a method that does not require any software/driver e.b. via BIOS.
- [Intel Power Maximizer](http://www.intel.com/performance/maximizer) (http://www.intel.com/performance/maximizer)

h3 Internet Latency "tweak" programs

- [TCPOptimizer \(speedguide.net\)](http://www.speedguide.net/files/TCPOptimizer.exe) (<http://www.speedguide.net/files/TCPOptimizer.exe>) *I do not recommend it*, tweak (manually) only parameters you [really need](https://github.com/CHEF-KOCH/GamingTweaks/issues/24) (<https://github.com/CHEF-KOCH/GamingTweaks/issues/24>).

h3 PC Latency measurement programs

- [LatencyMon \(resplendence.com\)](http://www.resplendence.com/latencymon) (<http://www.resplendence.com/latencymon>)

h3 Mouse "OC" programs

- [Mouse Overclocking Utility \(github.com\)](https://github.com/vadash/1000hz) (<https://github.com/vadash/1000hz>)

h3 Fix broken stereoscopic effects

- [3Dmigoto \(github.com\)](https://github.com/bo3b/3Dmigoto) (<https://github.com/bo3b/3Dmigoto>) - Chiri's DX11 wrapper to enable fixing broken stereoscopic effects.

h3 Power Management programs

- [ClockBlocker \(profiled AMD power-management control\) \(forums.guru3d.com\)](https://forums.guru3d.com/threads/clockblocker-profiled-amd-power-management-control.404465/) (<https://forums.guru3d.com/threads/clockblocker-profiled-amd-power-management-control.404465/>)

h3 Bios adjustment programs

- [BIOSTimingsDecoder](https://github.com/integralfx/BIOSTimingsDecoder) (<https://github.com/integralfx/BIOSTimingsDecoder>)

h3 Unpausable cutscenes

- [Game Pauser](https://madebyjase.com/game-pauser/) (<https://madebyjase.com/game-pauser/>) **(Do NOT use it for online games, due to anti-cheat reasons)**

h3 FOV calculator

- [Online FOV calculator](https://themetalmuncher.github.io/fov-calc/) (<https://themetalmuncher.github.io/fov-calc/>)

h3 GPU BIOS Updater tools

- [Video Bios Collection](https://www.techpowerup.com/vgabios/) (<https://www.techpowerup.com/vgabios/>)
- [AMDVbFlash](https://cdn.discordapp.com/attachments/529672908780339210/669884538335264788/AMDVbFlash_Win_v3.03.rar) (https://cdn.discordapp.com/attachments/529672908780339210/669884538335264788/AMDVbFlash_Win_v3.03.rar) also known as [AMD/ATI ATIFlash](https://www.techpowerup.com/download/ati-atiflash/) (<https://www.techpowerup.com/download/ati-atiflash/>) (*which is the old name*) + [Tutorial](https://www.gamersnexus.net/guides/3550-guide-how-to-flash-amd-gpu-vbios-rx-5600-xt) (<https://www.gamersnexus.net/guides/3550-guide-how-to-flash-amd-gpu-vbios-rx-5600-xt>)
- [nVidia NVFlash](https://www.techpowerup.com/download/nvidia-nvflash/) (<https://www.techpowerup.com/download/nvidia-nvflash/>) + [Tutorial](https://www.overclock.net/forum/69-nvidia/1523391-easy-nvflash-guide-pictures-gtx-970-980-a.html) (<https://www.overclock.net/forum/69-nvidia/1523391-easy-nvflash-guide-pictures-gtx-970-980-a.html>)

h2 Game streaming services

Provider	Comment
Amazon Game Studios (https://www.amazongames.com/)	//
Blade Shadow (https://shadow.tech/gben/discover)	Works with Mac, Android, iPhone, Ubuntu and Windows 10.
Google Stadia (https://stadia.com/)	Supported game list (https://store.google.com/de/product/stadia_games)
Twitch (https://www.twitch.tv/)	Planned for 2020
Parsec (https://parsecgaming.com/)	Play, watch, and share gaming with your friends.
GeForce Now (https://www.nvidia.com/en-gb/geforce/products/geforce-now/)	Play & Stream Android, PC, and GeForce NOW games.
Steam Link (https://store.steampowered.com/steamlink/about/) & Steam Remote Play (https://store.steampowered.com/remoteplay/)	Stream video, audio, and voice between players.
Moonlight (https://moonlight-stream.org/)	Play your PC games remotely. Setup guide (https://github.com/moonlight-stream/moonlight-docs/wiki/Setup-Guide).
Rainway (https://rainway.com/)	Launch all your PC games from one place or stream through any browser, free.
PlayStation Now (https://www.playstation.com/en-gb/get-help/help-library/services/playstation-now/playstation-now-on-pc/)	Stream PS2/PS3/PS4 games.
Xbox Game Streaming (xCloud) (https://www.xbox.com/en-US/xbox-game-streaming)	Stream Xbox games from the cloud to your Android phone or tablet.
Paperspace (https://www.paperspace.com/)	Fast cloud gaming in a web browser. Setup guide (https://blog.parsecgaming.com/setting-up-game-streaming-in-the-cloud-with-parsec-paperspace-just-got-a-lot)

Provider	Comment
	-easier-bec072732bc9).
Vortex Cloud Gaming (https://vortex.gg/)	Cloud Gaming for Android, PC and macOS.
Shadow (https://shadow.tech/)	\$35/month, provides cloud gaming.
Magenta Gaming (https://magenta-gaming.telekom-dienste.de/)	Telekom (ISP) only, requires 50 Mbit/s.

Virtual LAN Gaming

Provider	Comment
Hamachi (https://www.vpn.net/)	Hosted VPN service for LAN-like networks (Windows, Mac, Linux).
Tunngle (https://web.archive.org/web/20170606182108/http://www.tunngle.net/)	Freemium and closed source (last version 5.8.9, Jun 2017).
Evolve (Player.me) (https://player.me/)	Inbuilt LAN support for almost all popular LAN games. Free, Party + \$4/month.
GameRanger (https://www.gameranger.com/download/)	One of the best Hamachi alternatives. Free, Silver \$19.95/year, Gold \$39.95/year.
Wippien (http://www.wippien.com/download.php)	P2P connection between PCs. Free and open source.
NetOverNet (https://www.netovernet.com/)	Simple VPN emulator for private gaming sessions. Free, Basic \$5/month, Advanced \$10/month.
Radmin VPN (http://www.radmin-vpn.com/)	Secure VPN tunnel with no user/gamer limit.
FreeLAN (http://www.freelan.org/download.html)	Almost no lag. Completely open-source. Tutorial here (https://www.freelan.org/get_help.html).
P2PVPN (http://p2pvpn.org/)	Open source, developed in Java. Development stopped in 2010.
ZeroTier One (https://www.zerotier.com/download.shtml)	Open source, end-to-end encrypted virtual networks.
SoftEther VPN (https://www.softether.org/5-download)	Supports Windows, Mac, Linux, Solaris and FreeBSD.
WireGuard (https://wiki.archlinux.org/index.php/WireGuard)	Fast, modern VPN with state-of-the-art cryptography.

Do I need an internal sound card?

This question depends on multiple factors:

- Are you [audiophile](https://en.wikipedia.org/wiki/Audiophile) (https://en.wikipedia.org/wiki/Audiophile)?
- Do you use "professional" headsets or speaker equipment? (*see explanation "The OHM problem" below*)
- How old are you? A well-known fact is that the older you get the ["less you hear"](https://www.hear-it.org/younger-and-older-people-respond-differently-sound) (https://www.hear-it.org/younger-and-older-people-respond-differently-sound) and your tolerance (*overall*) for loud noise [decreases with age](https://www.eurekalert.org/pub_releases/1998-11/OU-TFLN-191198.php) (https://www.eurekalert.org/pub_releases/1998-11/OU-TFLN-191198.php). It's a [myth that older people have automatically ear "problems"](https://www.fastcompany.com/3064873/the-reason-why-old-people-have-bad-hearing-has-nothing-to-do-with-their-ears) (https://www.fastcompany.com/3064873/the-reason-why-old-people-have-bad-hearing-has-nothing-to-do-with-their-ears).

The OHM problem

Motherboards generally do not have enough power for complex speaker outputs, for example: - [Dolby Atmos](https://www.dolby.com/us/en/brands/dolby-atmos.html) (https://www.dolby.com/us/en/brands/dolby-atmos.html). Additionally, high-end headphones are typically rated at ~600 ohms, which is much more power than a motherboard audio IC can supply; therefore, an AMP/DAC would be required. With that being said, for headphones the ideal situation would be using an external AMP. For speakers the ideal situation is to run HDMI from your GPU into an AVR and then plug your speakers into the AVR.

CPU problems

Integrated audio solutions very often exclude a SPU which means your CPU is doing the entire work. The same goes for "audio effects", for example: - [Nahimic](https://www.msi.com/page/naahimic) (https://www.msi.com/page/naahimic) (*each vendor names his effect different, MSI calls it "Nahimic"*), if you enable "a lot" or even one sound effect you automatically put more "pressure" on your CPU unless the sound chip includes an "acceleration" for this within the SPU.

DAC & Drivers

If you think a sound card would improve your experience, you're mostly always better off purchasing a DAC because (*overall*) it has a better value to performance and is more flexible in terms of performance. Another benefit is that a DAC does not need any drivers and there are no "PCIE issues". Overall a DAC can deliver the same benefit but without running into *possible* driver problems.

h3 Competitive fps and the "footstep" problematic

Some say this is placebo effect, while others swear there is a benefit in using sound effects to "enhance" enemy footstep sound in e.g. CS:GO. From my experience this is a *matter of taste*, however if you want to enhance something you could use [Equalizer APO](https://sourceforge.net/projects/equalizerapo/) (<https://sourceforge.net/projects/equalizerapo/>), which can (*in real-time*) without much CPU pressure change the microphone, speakers/headset and correct sound frequencies.

h2 Conclusion

3.5 mm "gaming headsets" and other cheaper speakers aren't going to benefit you much with an internal or aftermarket sound card. A DAC has more "benefits" and is overall easier to handle. It's more flexible because there are limited driver and software problems.

You typically do not need a sound card unless:

- You have a dated CPU which can't handle the "extra pressure" for the sound processing sequence. The argument is generally weak these days, and because CPU's are so fast plus we have multiple cores this should not be a big argument anymore. Gamers generally always have "multi-core" CPU's.
- Sound effects are "cool" on paper but they can put extra pressure on your CPU, however modern games are typically designed to sound balanced (out-of-the-box) because game developers spend lots of time "optimizing" the sound for a maximum sound experience. Some older games do have "bad sounds" but even then, the community often provide "sound enhancement mods" to fix it, so there is no "pro argument" to use sound effects.
- Do not use speakers or headphones which "require" 600 Ohm; however, you can still can use the hardware with the onboard sound but it will sound worse: - ‘worse’ in context means noticeably lower volume with muddier bass.
- If you have speakers and want encoded audio like Dolby Atmos 5.1.4 or 7.1.4 then you should use an AVR. The problem here is that the independence and Ohm(s) are not "high" enough to handle this directly via an internal sound card to sound "powerful".

h2 Realtek specific

Warning: Sound effects will slightly increase the [DPC latency](https://superuser.com/questions/1206472/dpc-latency-issue-caused-by-realtek-controller-driver) (<https://superuser.com/questions/1206472/dpc-latency-issue-caused-by-realtek-controller-driver>), which in turn could cause other problems like audio and video desynchronization or slow network speed. Sound enhancements could also have an adverse effect; it may sound better with your music, but could harm your audio positioning in games.

h3 Windows 10 specific Realtek settings

- In Windows 10 under "Speaker Configuration" select the number of speakers that you actually have. For a "5.1" audio system select "5.1" Speaker. For a "2.1" audio system select "Stereo". Selecting more channels than the number of speakers you have connected won't enable you to hear more sounds or hear the footsteps and positioning of enemies better in games.
- Set the default playback format to "24Bit", "44100 Hz" this is generally good for most media (games, movies, music), while video DVDs use "48 kHz". Increasing the actual "Sample Rate" without having a source that's sampled at a higher rate won't improve the overall sound quality - in fact - it will degrade quality instead. Having the bit-depth set to 24bit will slightly improve audio quality. [You can find detailed information on audio bit-depth here](http://wiki.jriver.com/index.php/Audio_Bitdepth) (http://wiki.jriver.com/index.php/Audio_Bitdepth)
- If your Realtek Audio solution has a dedicated headphone amplifier, set the soundcards output impedance/gain (Ω = ohm) to a value close to the headphones input impedance. Do not set the impedance much higher as it may damage the headphones by outputting too much power.
- In case you like to emulate 8 channels surround on 2 channel headphones or speakers you need to right-click on Windows Volume tray icon-> Spatial Sounds-> select “Windows Sonic for Headphones”. Optionally, you can also try "Dolby Access", "DTS Sound Unbound" or "Dolby Atmos for Headphones" options. Once enabled it will automatically (*in the background*) change playback audio bit depth to 16bit with a sample rate of 44100 Hz. When disabled Windows 10 will set the playback audio bit depth to 24bit and sample rate to 48000 Hz (*this is by design*).
- Optional: Set each individual application's volume to ~80-90% in order to avoid "fluctuating volume" issue. Issue regarding setting volume to 100%. However, on newer Windows 10 builds (1809+) with the newer Realtek drivers this is not needed anymore.

h3 Microphone

- Set the microphone and/or ‘Line In’ default format to "24bit", "44100 Hz". The 44.1 kHz (44100 Hz) sample rate is the one used by most media (games, movies, music), while video DVDs use 48 kHz. Having the bit-depth set to 24 bit will slightly improve the recording audio quality, most USB mics and 3.5 mm mics (*unless it's a professional device*) still using 44.1 kHz and there is no "benefit" nor "enhancement" setting it to 48.0 kHz (this also applies to recoding software like OBS Studio).

- If you’re using speakers and want to reduce the speaker output from being picked up by the microphone then enable Acoustic Echo Cancellation option under “Microphone Effects”.

Common Tips

- Always select 2/2.1 speakers for your headphones if you want to utilize the full dynamic range, unless you have a real (not virtual) 7.1 headset. If set incorrectly you may experience sound disorientation, FPS especially benefit from a proper setup.
- Since the Windows April Update (Build 1803) you don’t need any external audio routing software in order to send the audio through a specific device or channel. This is [explained in detail over here](https://www.groovypost.com/howto/new-sound-settings-windows-10-1803-april-update/) (https://www.groovypost.com/howto/new-sound-settings-windows-10-1803-april-update/).
- Old Realtek drivers (up to Win 10 Redstone 2) were packaged in 200+ MB files, this has changed and improved since Windows RS 3 which means the Universal Audio Drivers (UAD) now come in smaller packages which weigh 20-40 MB max, depending which audio effects your OEM supports and can additionally install like ASIO, DSP, Creative or Windows Effects. However, normal non-OEM packages (the ones which are offered by Windows Update or Update Catalogue) usually exclude them (for no reason). So, you can 'tweak' your latency via ASIO and 'pimp' your effects by installing them additionally. I upload them separately under **Realtek ASIO, effects etc** so you can drop them into the installation folder or right click on the .inf and 'Install' it in order to unlock it.
- Universal Audio driver (UAD) and High Definition Drivers (HDA) drivers, including the latest UWP apps are available [here](https://github.com/alanfox2000/realtek-universal-audio-driver) (https://github.com/alanfox2000/realtek-universal-audio-driver).
- Enable the audio output/playback to 24bit in media applications like VLC, MPC-BE/HC etc., this will slightly improve audio quality if the output material has a depth of 24bit (*which is the default today*).

In-Game Tips

Set the sound quality to **Medium** or **High** (if there is such an option) depending on the importance of the sound quality for that particular game. This setting affects CPU usage. If you have a slow CPU you should select ‘Medium’, if you have a fast CPU you should select ‘High’. Setting the option to ‘High’ on a dated CPU may cause too much strain on your CPU and in turn can affect the FPS of that particular game.

EqualizerAPO Tutorials

- [EqualizerAPO + ReaPlugs VSTs \(Mic processing without virtual cables\)](https://www.youtube.com/watch?v=J3fBx2ftaBs) (https://www.youtube.com/watch?v=J3fBx2ftaBs)
- [Fixing Microphone Reverb with Equalizer APO](https://www.youtube.com/watch?v=eySlFVYGk8E) (https://www.youtube.com/watch?v=eySlFVYGk8E)

Modded sound drivers from DanielK, PAX & Co.

In some situations a modded sound driver or all-in-one packages can has its benefits, e.g. when the original driver causes problems or does not include features like DolbyAtmos. If you need such features or want to test those tweaked driver/packages, then there are two great resources (*see links below*).

- [Daniel_K](https://danielkawakami.blogspot.com/) (https://danielkawakami.blogspot.com/) is a well-known sound card driver modder mainly for Realtek/Creative/Asus cards. Since the end of 2017 he did not updated his homepage nor driver packages.
- [PAX](https://www.hardwareheaven.com/community/forums/pax-drivers.174/) (https://www.hardwareheaven.com/community/forums/pax-drivers.174/) is a still active and known for his Creativte/Asus sound packages/driver mods and tweaks. Some of his packages/mods are based in Daniel_K's work.
- [The Ultimate Realtek HD Audio Driver Mod for Windows 10](https://www.techpowerup.com/forums/threads/the-ultimate-realtek-hd-audio-driver-mod-for-windows-10.232921/page-112) (https://www.techpowerup.com/forums/threads/the-ultimate-realtek-hd-audio-driver-mod-for-windows-10.232921/page-112) is a thread on TechPowerUP around modding Realtek specific drivers to get Dolby, DTS, DTS:X 2,0 working. The thread provides All-in-One (AiO) packages, which you cn install on Windows 10 to get more sound effects and features. The member [Shibajee Roy](https://github.com/shibajee?tab=repositories) (https://github.com/shibajee?tab=repositories) extracted everything from the AIO's an created repositories for each individual brand and sound effect, which makes the *Realtek modding* overall easier.

SSD

Setting **fsutil behavior set disablelastaccess 1** still needs to be set manually. There is research on this available [here](https://translate.google.com/translate?hl=en&sl=ja&u=http://blog.livedoor.jp/nichepcgamer/archives/1071061265.html&prev=search) (https://translate.google.com/translate?hl=en&sl=ja&u=http://blog.livedoor.jp/nichepcgamer/archives/1071061265.html&prev=search).

TRIM

Since Windows 10, the OS usually checks the [TRIM](https://en.wikipedia.org/wiki/Trim_(computing)) (https://en.wikipedia.org/wiki/Trim_(computing)) status on his own, but in order to verify it simply run **fsutil behavior query DisableDeleteNotify** . If TRIM is functional, **DisableDeleteNotify=0** will be returned; otherwise, **DisableDeleteNotify=1** will be returned.

Disk Defragmentation

You don't need to disable the internal Disk defragmentation program, it detects which kind of storage you use and start/stop the progress on it's own. There was a Bug with >1607 Builds to detect it, but was fixed later by Microsoft. There is no benefit disabling the schedule (Hibernation) via **powercfg -h off** .

Enable Write Caching for SSDs

Write caching can theoretically improve performance of both SSDs & HDDs. It's usually enabled by default.

h3 Write-cache Buffer Flushing

Using this is not recommended in case of a power failure you might loose your data.

h3 Disable Prefetch and Superfetch

SuperFetch (SysMain) indexes your SSD/HDD which means when you search for certain things, the results are showing up faster (in theory). If you use search utilities like [Everything](https://www.voidtools.com/downloads/) (https://www.voidtools.com/downloads/) you don't need such function.

h3 Integrated Telemetry into games (e.g. DRM, Redshell & Co.) and Game Clients

Some games (not all) including copyright systems like Denuvo (DRM) checks or [Redshell](https://docs.google.com/spreadsheets/d/e/2PACX-1vQz1d2jf15nHZE8GaRDAWCVMWuYkhip_cwkDUD3fo9dn0EiDRG3crtNXNhPESz8ZLL2KVDULnm9D-VB/pubhtml) (https://docs.google.com/spreadsheets/d/e/2PACX-1vQz1d2jf15nHZE8GaRDAWCVMWuYkhip_cwkDUD3fo9dn0EiDRG3crtNXNhPESz8ZLL2KVDULnm9D-VB/pubhtml) in order to spy & log the gamers activity which has an negative impact of the game performance itself ([longer loading times](https://www.youtube.com/watch?v=ByfLg9wGB4o) (https://www.youtube.com/watch?v=ByfLg9wGB4o), FPS drops or even crashes). A filter-list to prevent such stuff can be found [here](https://github.com/CHEF-KOCH/CKs-Filter-List) (https://github.com/CHEF-KOCH/CKs-Filter-List).

Keep in mind that [telemetry](https://en.wikipedia.org/wiki/Telemetry) (https://en.wikipedia.org/wiki/Telemetry) has (in normal circumstances) no effect (or less) on the gaming performance (if it was proper implemented) and you should *consider to leave it enabled* in order to help the game developers (e.g. for crash reports or to improve the anti-cheat mechanism) however, such systems scanning your entire OS and this might invade your privacy. I highly recommend that you read the terms of service before you opt-in.

The biggest impact on the game performance itself is done by the copyright-protection system and its implementation and *can have* a negative impact which results in longer loading times or *small* FPS drops. The rumour that Denuvo cripples your performance is not *entirely* true, it's depending on several factors in most cases you [lose around 2-3 FPS](https://www.extremetech.com/gaming/282924-denuvo-really-does-cripple-pc-gaming-performance) (https://www.extremetech.com/gaming/282924-denuvo-really-does-cripple-pc-gaming-performance) (which is considerable not "crippled" - depending how you see it).

h4 My advice in general is:

- Use [GOG](https://www.gog.com/) (https://www.gog.com/) releases because they are DRM free, some Epic Game Store and Steam Store Games are also DRM free but the Client itself is not.
- Play cracked versions (*possible malware risk*) or emulate them in a VM/Wine/Sandbox. You still can support the game developer by buying the game, just don't install it via *official ways*.
- Check if there are *confirmed* and *clean* cracks from [trusted groups](https://crackwatch.com/) (https://crackwatch.com/) which avoids losing some performance via VM/Sandbox (as mentioned above) but the risk is still there (!) that you get infected (via re-packs, drive-by, Malvertising etc). *It is up to you*.
- Check if you [need the Game Client](https://www.resetera.com/threads/developing-epic-games-launcher-appears-to-collect-your-steam-friends-play-history-epic-responds-see-op.105385/) (https://www.resetera.com/threads/developing-epic-games-launcher-appears-to-collect-your-steam-friends-play-history-epic-responds-see-op.105385/) and read the privacy policy before you install it on your OS!
- Manually removing the analytics code requires to manipulate the game client and it's files and is *dangerous* because that might result in a ban, the game client should provide an opt-out toggle for such tracking. A better method is to block the request with a firewall/hosts file.

h2 In which cases Telemetry has no impact on the OS or gaming performance

- Depending how well it was [implemented \(example\)](https://github.com/matthewhorridge/telemetry) (https://github.com/matthewhorridge/telemetry), telemetry (the running processes) never uses much resources (*depending how you define "much"*).
- Telemetry can run in the background or even without user consent, the best scenario would be if you can [manually control it opt-in/opt-out](https://blogs.windows.com/buildingapps/2014/03/20/instrumenting-your-app-for-telemetry-and-analytics/) (https://blogs.windows.com/buildingapps/2014/03/20/instrumenting-your-app-for-telemetry-and-analytics/) and submit only telemetry when you want it
- Checks could be integrated to scan the current running applications and if they are currently in fullscreen or not (e.g games)
- The Telemetry and his task schedule can be configured to run only on specific circumstances e.g. if the PC is idle
- Depending on how big the submitted data are or how small they are you will never notice it (not in-game and not while you use other applications)

h1 Reset_winsock

```
> rem Reset the entire Winsock/firewall

rem Based on:
rem https://support.microsoft.com/en-us/help/10741/windows-10-fix-network-connection-issues (https://support.microsoft.com/en-us/help/10741/windows-10-fix-network-connection-issues)
```

```
rem DOCS
rem advfirewall - technet.microsoft.com/en-us/library/cc771046\(v=ws.10\).aspx
rem arp - technet.microsoft.com/en-us/library/cc940107.aspx
rem ipconfig - technet.microsoft.com/en-us/library/bb490921.aspx
rem nbstat - technet.microsoft.com/en-us/library/bb490938.aspx
rem netsh - technet.microsoft.com/en-us/library/cc770948\(v=ws.10\).aspx
rem route - technet.microsoft.com/en-us/library/bb490991.aspx

rem Set all needed services back to start/auto
reg add "HKLM\System\CurrentControlSet\Services\BFE" /v "Start" /t REG_DWORD /d "2" /f
reg add "HKLM\System\CurrentControlSet\Services\Dnscache" /v "Start" /t REG_DWORD /d "2" /f
reg add "HKLM\System\CurrentControlSet\Services\MpsSvc" /v "Start" /t REG_DWORD /d "2" /f
reg add "HKLM\System\CurrentControlSet\Services\WinHttpAutoProxySvc" /v "Start" /t REG_DWORD /d "3" /f

rem Default services
sc config Dhcp start= auto
sc config DPS start= auto
sc config lmhosts start= auto
sc config NlaSvc start= auto
sc config nsi start= auto
sc config RmSvc start= auto
sc config Wcmsvc start= auto
sc config WdiServiceHost start= demand
sc config Winmgmt start= auto

sc config NcbService start= demand
sc config Netman start= demand
sc config netprofm start= demand
sc config WlanSvc start= auto
sc config WwanSvc start= demand

net start Dhcp
net start DPS
net start NlaSvc
net start nsi
net start RmSvc
net start Wcmsvc

rem Disable netadapter with index number 0-5 (ipconfig /release)
wmic path win32_networkadapter where index=0 call disable
wmic path win32_networkadapter where index=1 call disable
wmic path win32_networkadapter where index=2 call disable
wmic path win32_networkadapter where index=3 call disable
wmic path win32_networkadapter where index=4 call disable
wmic path win32_networkadapter where index=5 call disable

rem Timeout to let the network adapter recover
timeout 6

rem Enable adapter with index number 0-5 (ipconfig /renew)
wmic path win32_networkadapter where index=0 call enable
wmic path win32_networkadapter where index=1 call enable
wmic path win32_networkadapter where index=2 call enable
wmic path win32_networkadapter where index=3 call enable
wmic path win32_networkadapter where index=4 call enable
wmic path win32_networkadapter where index=5 call enable

rem Reset winsock, adapters and firewall (ignore the errors)
arp -d *
route -f
nbstat -R
nbstat -RR
netsh advfirewall reset

netcfg -d
```

```
netsh winsock reset
netsh int 6to4 reset all
netsh int httpstunnel reset all
netsh int ip reset
netsh int isatap reset all
netsh int portproxy reset all
netsh int tcp reset all
netsh int teredo reset all
ipconfig /release
ipconfig /renew
ipconfig /flushdns

rem OPTIONAL: Force restart in 1 minute
rem shutdown /r /t 60

rem Take ownership of the following key in registry if you get "Access denied" message, then set Permissions for the current user to "Allow Full Control" or try to
run the bat in safe mode
rem HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Control\Nsi\{eb004a00-9b1a-11d4-9123-0050047759bc}\26
rem Default entries are 10, 11, 12, 16, 18, 26, 30, 4, 6, 7

rem To disable restart, type - shutdown /a

pause
```